

LYRA: GENERATIVE 3D SCENE RECONSTRUCTION VIA VIDEO DIFFUSION MODEL SELF-DISTILLATION

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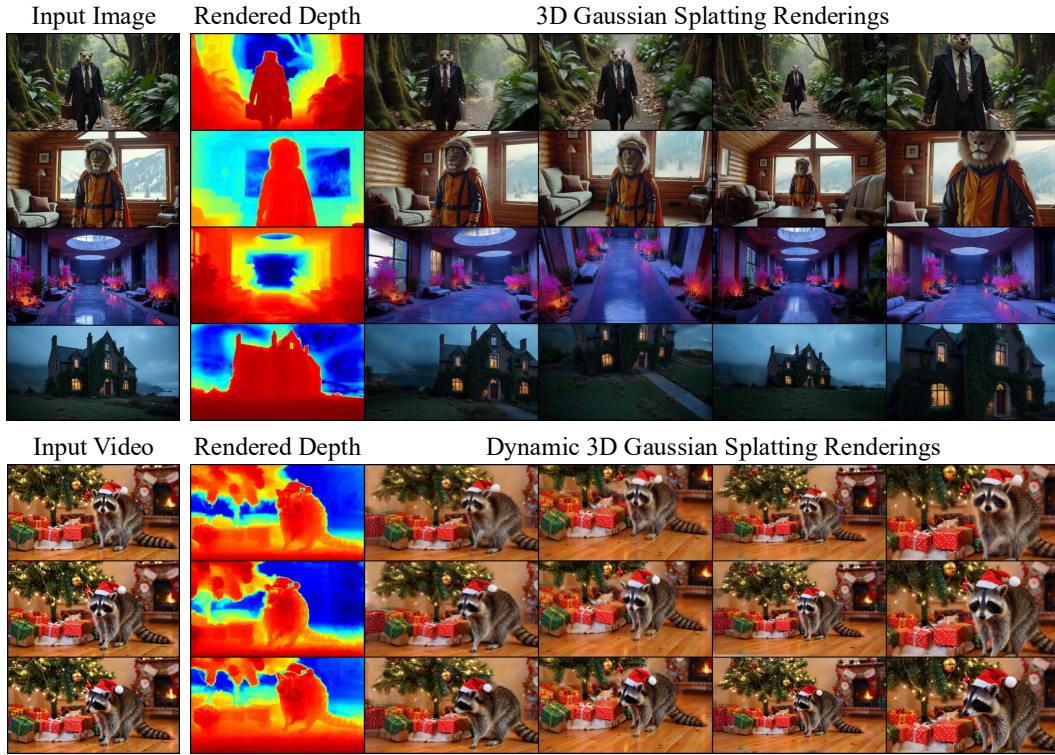


Figure 1: **Feed-Forward 3D and 4D Scene Generation.** From a single image (top), *Lyra* infers a 3D Gaussian Splatting (3DGS) representation in a feed-forward manner by self-distilling a video diffusion model, without relying on real-world multi-view data. Given a video input (bottom), *Lyra* infers a dynamic 3DGS that provides interactive control over both time (rows) and viewpoint (columns).

ABSTRACT

The scalable generation of virtual environments is increasingly consequential for applications that range from entertainment media to physical-AI domains, including robotics, autonomous driving, and industrial automation. Contemporary learning-based 3D reconstruction methods, however, typically presuppose captured real-world multi-view observations, a form of supervision that is often expensive, logistically constrained, or altogether unavailable. Although recent video diffusion models exhibit striking generative and imaginative capacity, their intrinsically two-dimensional outputs restrict their utility in simulation settings where robots or embodied agents must navigate, interact with, and reason over geometrically persistent environments. In this paper, we introduce a self-distillation framework designed to transfer the implicit 3D knowledge encoded by video diffusion models into an explicit 3D Gaussian Splatting (3DGS) representation, thereby removing the need for multi-view training data. Concretely, we augment the conventional RGB decoder with a 3DGS decoder, whose predictions are supervised by the RGB decoder’s outputs. Under this formulation, the 3DGS decoder can be trained

exclusively from synthetic data produced by video diffusion models. At inference time, the resulting model synthesizes 3D scenes from either a text prompt or a single image while supporting real-time rendering. The same framework further extends to dynamic 3D scene generation from monocular input video. Experiments demonstrate that our approach attains state-of-the-art performance for both static and dynamic 3D scene generation.

1 INTRODUCTION

The scalable creation of high-quality 3D environments remains a central and persistent challenge in computer vision and graphics, with direct implications for film-making, VR/AR, and physical AI systems that require closed-loop simulation. Across these application domains, utility depends not merely on photorealistic imagery, but on explicit 3D representations that permit real-time rendering, physical interaction, and coherent synthesis from multiple viewpoints.

Recent progress in neural 3D reconstruction (Mildenhall et al., 2020; Kerbl et al., 2023; Moenne-Loccoz et al., 2024) has made it possible to recover such representations from posed image collections; nevertheless, this paradigm remains difficult to scale because it depends on accurate camera poses and high-quality visual observations. The difficulty is amplified in dynamic scenes, where reliable reconstruction commonly requires synchronized multi-camera acquisition (Li et al., 2022). Moreover, reconstruction methods are intrinsically bounded by the content actually observed in the input views and therefore cannot reliably extrapolate beyond their visual evidence. Although feed-forward reconstruction models (Zhang et al., 2024e; Ren et al., 2024c) partially mitigate these constraints, they introduce a different bottleneck: the limited availability of diverse, large-scale 3D training data, which in turn impairs out-of-distribution generalization.

Video diffusion models (Cosmos, 2025; Wan, 2025) provide a compelling alternative source of supervision. Because they are trained on massive internet-scale video corpora, these models achieve strong visual fidelity and broad generalization across heterogeneous environments; in learning from real-world videos with camera trajectories that span handheld motion, cinematic pans, and drone footage, they implicitly acquire cues about the underlying 3D world without requiring curated multi-view datasets. Unlike reconstruction approaches (Charatan et al., 2024), generative models can also synthesize plausible content outside the initially visible field of view. Their limitation, however, is that they produce 2D frame sequences rather than explicit 3D representations, which restricts their applicability in simulation and other downstream settings that require geometric consistency, long-horizon coherence, and physical interaction. Encouragingly, recent work (Wang et al., 2024e; Ren et al., 2025) has shown that video models can be adapted for explicit camera control using comparatively small posed datasets, thereby transforming posed video generation into a useful source of both input and supervision for 3D reconstruction models.

Motivated by this observation, we connect the paradigms of reconstruction and generation through what we term *generative 3D scene reconstruction*. We introduce *Lyra*, a method that produces explicit 3D environments from the latent representations of video diffusion models in a single forward pass. Crucially, *Lyra* is trained through a self-distillation framework (Fig. 2) in which a 3D Gaussian Splatting (3DGS) decoder operates directly in the latent space of a video diffusion model. Given a single input image or video, we sample a camera trajectory as conditioning, denoise the corresponding video latent, and decode it through two parallel branches: a standard RGB decoder that synthesizes a video sequence, and a 3DGS decoder that predicts an explicit 3D representation. These branches instantiate a teacher–student procedure in which the RGB branch supervises the 3DGS branch. To expand viewpoint coverage, we sample multiple camera trajectories, generate multiple video latents, and train the 3DGS decoder to integrate information across them while reducing long-term and multi-view inconsistencies.

This generative reconstruction framework offers three principal advantages: (i) it enables the generation of large-scale synthetic environments across diverse scenarios directly from video diffusion models, eliminating reliance on real-world multi-view capture; (ii) by operating in the latent space of a video model, it permits efficient processing of many views without incurring the substantial memory costs associated with pixel-space feed-forward reconstruction; and (iii) its explicit 3DGS output imposes geometric consistency and yields representations that can be used directly in downstream settings such as physical simulation. Consequently, at inference time, *Lyra* generates high-quality

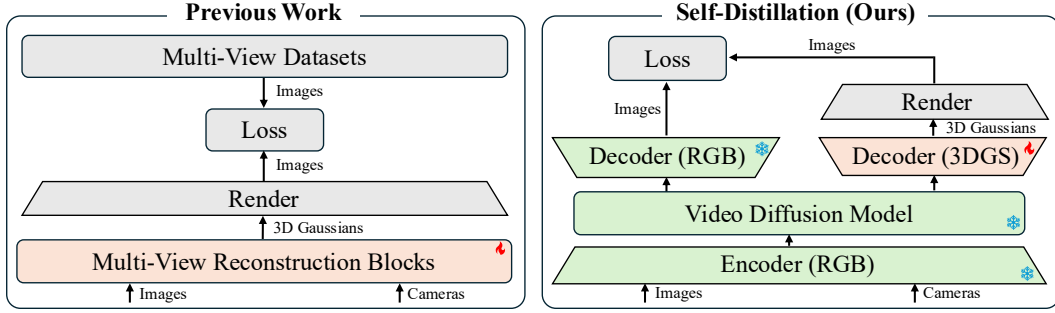


Figure 2: **Self-distillation framework of *Lyra***. Prior work (left) (Szymanowicz et al., 2025b) trains multi-view reconstruction blocks on real-world datasets whose diversity is limited (Zhou et al., 2018; Ling et al., 2024b). In contrast, our self-distillation framework (right) targets generative 3D scene reconstruction by using a pre-trained camera-controlled video diffusion model and its RGB decoder outputs as the teacher that supervises renderings from the 3DGS decoder student. Because the video model is pre-trained on diverse 2D video data, it supplies broad multi-view supervision without captured multi-view 3D data.

3D Gaussian scenes from monocular input that support real-time rendering, *without requiring any additional optimization or post-processing*.

We further generalize the self-distillation framework to dynamic 4D generation from monocular video. In this setting, the video model functions as a teacher that supplies space–time supervision, while the student learns time-conditioned 3DGS representations that enable novel-view synthesis of dynamic scenes.

Overall, our work makes the following contributions:

- We introduce a self-distillation framework that trains a 3DGS student decoder using a pre-trained camera-controlled video diffusion model as the teacher, eliminating the need for captured multi-view real-world data.
- Our framework extends to dynamic scenes to support 4D reconstruction from a monocular video.
- Our model generalizes across diverse scenes, achieving state-of-the-art results in single-image 3D scene generation and single-video 4D scene generation.

2 RELATED WORK

Multi-view image generation. Early work on multi-view image generation (Watson et al., 2023; Liu et al., 2023; Shi et al., 2024; Wang & Shi, 2023) primarily addressed object-centric settings in which background structure was absent or simplified. Subsequent methods (Sargent et al., 2024; Tang et al., 2023b; Schneider et al., 2025) extended multi-view image generation toward scene-scale synthesis. CAT3D (Gao et al., 2024b), for example, adapts a pre-trained image diffusion model to generate multiple views from either generated or real image inputs; in a second stage, those synthesized views are reconstructed into an explicit 3D representation, such as NeRF (Mildenhall et al., 2020) or 3DGS (Kerbl et al., 2023). Related methods have further generalized this strategy to dynamic 3D scenes by generating multiple viewpoints with a space-time diffusion model (Watson et al., 2024; Wu et al., 2025b; Kuang et al., 2024; Wang et al., 2025a). Despite their impressive visual results, these approaches still require an expensive per-scene optimization stage to ground generated imagery in an explicit 3D representation, and this computation is not amortized across scenes. In contrast, rather than first producing multi-view images and subsequently reconstructing them through optimization, our objective is to *directly* generate a 3D scene from text or a single image.

Camera-conditioned video generation. A substantial body of recent work has investigated the fine-tuning of video diffusion models for 3D camera control. MotionCtrl (Wang et al., 2024e) pioneered this direction by conditioning pre-trained video models on camera poses. Later approaches (He et al., 2024a; Xu et al., 2024c; Bahmani et al., 2025a;b) represent cameras with Plücker coordinates to provide pixel-wise conditioning, while another line of work (Hu et al., 2024; Hou et al., 2024; Zhang et al., 2024b) studies training-free camera control for video diffusion models. More recently, SynCamMaster (Bai et al., 2025b) and ReCamMaster (Bai et al., 2025a) have demonstrated high-quality camera-controlled video synthesis using large-scale synthetic training data. Although these models produce compelling controlled videos, their outputs remain ungrounded in an explicit 3D

representation. Our work is therefore orthogonal to this research thread: we address feed-forward 3D generation from camera-controlled video models, enabling novel-view rendering *after* generation. Concretely, we build on GEN3C (Ren et al., 2025), a recent camera-controlled video diffusion model, and deploy it as a teacher in a self-distillation framework for 3D reconstruction.

Feed-forward 3D models. Early feed-forward 3D reconstruction methods also tended to focus on object-centric scenes without backgrounds, directly predicting either a NeRF (Hong et al., 2024; Li et al., 2024b) or 3DGS (Tang et al., 2024a; Szymanowicz et al., 2024) from text or image input. More recent methods (Charatan et al., 2024; Zhang et al., 2024e; Szymanowicz et al., 2025a; Ren et al., 2024c) address scene-level 3D reconstruction, commonly by regressing per-pixel 3D Gaussians from one or more input images. However, most scene-level feed-forward reconstruction models remain tied to their training distributions, such as RealEstate10K (Zhou et al., 2018), and consequently generalize only weakly to generated or otherwise out-of-distribution scenes. Recent methods have begun extending feed-forward reconstruction toward scene generation: Bolt3D (Szymanowicz et al., 2025b) trains a pointmap (Wang et al., 2024b) autoencoder to generate pointmaps alongside multi-view images for feed-forward 3D reconstruction, while the closely related Wonderland (Liang et al., 2025a) uses a camera-controlled video model and a feed-forward network to produce 3D Gaussians. Unlike these works, we use a camera-controlled video model as a teacher in a self-distillation framework that trains the student 3D model *without requiring real-world multi-view data*. Importantly, we further show that the same framework admits a simple extension to feed-forward 4D scene generation, a task that has not previously been explored in this setting.

3 SELF-DISTILLATION USING VIDEO DIFFUSION MODELS

Our central idea is to distill the implicit 3D knowledge embedded within video diffusion models into an explicit 3DGS decoder that can produce high-quality 3D representations. To accomplish this, we instantiate a teacher–student framework in which the video diffusion model, acting as the teacher, generates RGB videos that supervise a 3DGS decoder, acting as the student, that operates in the same latent space, as illustrated in Fig. 2. This section first describes the video diffusion backbone (Sec. 3.1) and then explains how it supplies self-distillation signals for the 3DGS decoder (Sec. 3.2). We subsequently detail the architectural design of the 3DGS decoder in Sec. 4 and specify the minimal modifications required to extend the pipeline to dynamic 3D scenes in Sec. 5.

3.1 BACKGROUND: CAMERA-CONTROLLED AND 3D-CONSISTENT VIDEO DIFFUSION

Because the 3DGS decoder is supervised exclusively by the camera-controlled video diffusion model, the success of our framework depends strongly on the teacher’s 3D consistency. We therefore build on GEN3C (Ren et al., 2025), a recent camera-conditioned video diffusion model, and summarize below the design choices in GEN3C that are most relevant to our formulation.

Spatiotemporal 3D cache. To enhance video consistency and camera-control precision, GEN3C constructs a spatiotemporal 3D cache $\{\mathbf{P}^{t,v}\}$ from input image(s) or videos, where each $\mathbf{P}^{t,v}$ is a colored point cloud obtained by unprojecting the depth estimation (Wang et al., 2024a) of an RGB image captured from camera viewpoint v at time t . The cache is arranged as an $L \times V$ array, with L denoting the temporal length in frames and V denoting the number of camera views.

Rendering and structured guidance. GEN3C uses this cache by rendering each point cloud from arbitrary prescribed camera poses, producing RGB images $\mathbf{I}^{t,v}$ and disocclusion masks $\mathbf{M}^{t,v}$ according to $(\mathbf{I}^{t,v}, \mathbf{M}^{t,v}) = \mathcal{R}(\mathbf{P}^{t,v}, \mathbf{C}^t)$, where $\mathcal{R}$ is the rendering function that projects the 3D point cloud $\mathbf{P}^{t,v}$ onto the 2D camera plane under camera pose $\mathbf{C}^t$ at time t . Given a camera-pose sequence $\mathbf{C} = (\mathbf{C}^1, \dots, \mathbf{C}^L)$, rendering every cache element yields V videos of length L , which are then organized as image sequences $\mathbf{I}^v \in \mathbb{R}^{L \times 3 \times H \times W}$ and mask sequences $\mathbf{M}^v \in \mathbb{R}^{L \times 1 \times H \times W}$ for each view v . The disocclusion masks identify regions that the video diffusion model must complete, while the rendered images provide structured visual guidance for subsequent video generation.

Video variational autoencoder (VAE). Diffusion-based video generation models commonly use a video variational autoencoder (VAE) (Kingma & Welling, 2013; Rombach et al., 2022) to compress videos into a lower-dimensional latent space, thereby improving both training and inference efficiency. Given a RGB video $\mathbf{I} \in \mathbb{R}^{L \times 3 \times H \times W}$, a pre-trained VAE encoder $\mathcal{E}$ maps the video into the latent representation $\mathbf{z} = \mathcal{E}(\mathbf{I}) \in \mathbb{R}^{L' \times C \times h \times w}$. Training and inference of the diffusion model are then conducted in this latent space. The final video $\hat{\mathbf{I}} = \mathcal{D}_{rgb}(\mathbf{z})$ is recovered with a pre-trained VAE

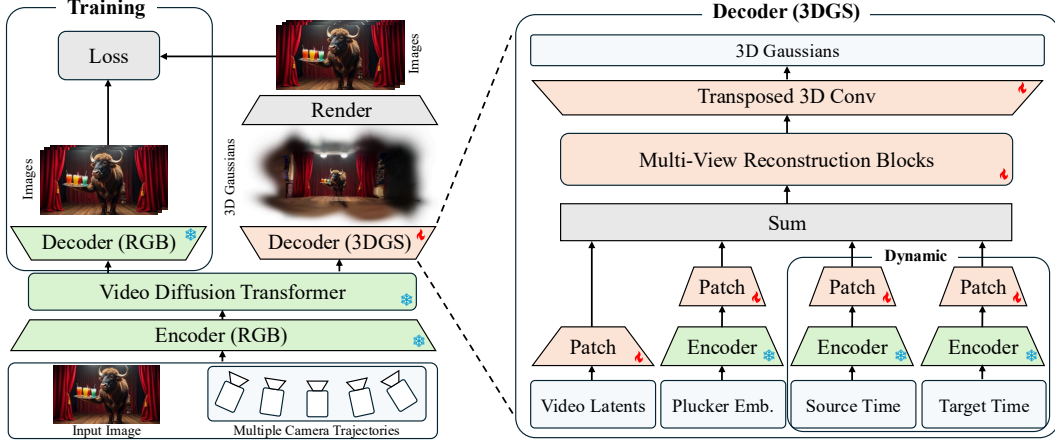


Figure 4: **3D Generative Reconstruction Framework.** Our pipeline builds on a camera-controlled video diffusion model (Ren et al., 2025) pre-trained on large-scale data. We train a 3D Gaussian Splatting (3DGS) decoder by aligning 2D renderings of generated 3DGS scenes with RGB-decoded generations from the frozen video model, updating only the 3DGS decoder while keeping the pre-trained autoencoder and diffusion model fixed. At inference time, the 3DGS decoder is used directly, so the RGB decoder is no longer required. Time conditioning inside the 3DGS decoder provides a straightforward extension from static to dynamic 3D scene generation.

decoder $\mathcal{D}_{rgb}$. In this paper, we adopt the pre-trained GEN3C (Ren et al., 2025; Cosmos, 2025) model; specifically, the latent channel dimension is $C = 16$, the temporal dimension is $L' = (L - 1)/\tau + 1$, and the spatial dimensions are $h = H/\sigma$ and $w = W/\sigma$, where the temporal compression factor is $\tau = 8$ and the spatial compression factor is $\sigma = 8$.

3.2 SELF-DISTILLATION

We train the 3DGS decoder $\mathcal{D}_s$ as a student in a teacher–student paradigm, with the camera-controlled video diffusion model $\mathcal{V}$ serving as the teacher. To obtain diverse supervision, we curate a large-scale collection of text prompts with large language models (OpenAI, 2025; Bai et al., 2023), synthesize images I with an image diffusion model (Black Forest Labs, 2024), and expand those images into multi-view sequences using GEN3C (Ren et al., 2025).

Teacher–student setup. Given an input image I and a sampled camera trajectory $\{\mathbf{C}^t\}_{t=1}^L$, the video diffusion model $\mathcal{V}$ produces a denoised video latent $\mathbf{z} = \mathcal{V}(I, \{\mathbf{C}^t\}_{t=1}^L)$. This latent $\mathbf{z}$ is decoded through two coordinated branches: the pre-trained RGB decoder $\mathcal{D}_{rgb}$ generates video frames $\mathbf{I}_{\mathcal{D}_{rgb}} = \mathcal{D}_{rgb}(\mathbf{z})$, whereas the 3DGS decoder $\mathcal{D}_s$ outputs explicit 3D Gaussians $\mathbf{G}$. Rendered views from the 3DGS (*student*) are given by $\mathbf{I}_{\mathcal{D}_s} = \text{Render}(\mathbf{G}, \{\mathbf{C}^t\}_{t=1}^L)$ and are optimized to match the RGB frames $\mathbf{I}_{\mathcal{D}_{rgb}}$ produced by the *teacher*, thereby closing the self-distillation loop.

Multi-trajectory supervision. To expand viewpoint coverage around each input image, we sample $V = 6$ camera trajectories, as shown in Fig. 3. For each trajectory v , we construct a spatiotemporal cache $\{\mathbf{P}^{t,v}\}_{t=1}^L$ together with $L = 121$ camera poses $\{\mathbf{C}^{t,v}\}_{t=1}^L$. Passing these caches through the video diffusion model $\mathcal{V}$ yields latents $\mathbf{z}^v$, which the teacher decodes into RGB frames $\mathbf{I}_{\mathcal{D}_{rgb}}^v = \mathcal{D}_{rgb}(\mathbf{z}^v)$. The 3DGS decoder $\mathcal{D}_s$ is then trained to fuse the multiple $\mathbf{z}^v$ into coherent Gaussians $\mathbf{G}$ while completing disoccluded regions. Through this self-distillation design, $\mathcal{D}_s$ learns entirely from synthetic supervision produced by $\mathcal{V}$; operating in latent space enables efficient aggregation across trajectories, whereas the explicit Gaussian output $\mathbf{G}$ promotes geometric consistency and supports downstream simulation and real-time rendering.

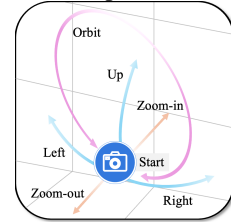


Figure 3: Six sampled camera trajectories selected to maximize viewpoint coverage.

4 FEED-FORWARD RECONSTRUCTION FROM MULTI-VIEW VIDEO LATENTS

Our feed-forward 3DGS decoder $\mathcal{D}_s$ converts the synthesized multi-view latents generated by the video model $\mathcal{V}$ into an explicit 3D representation that can be rendered from arbitrary viewpoints. We adopt 3D Gaussian Splatting (3DGS) (Kerbl et al., 2023) because it provides an explicit scene representation together with fast rendering. The complete 3DGS decoder is visualized in Fig. 4 (right).

4.1 3DGS DECODER

Scaling Latent-Based 3D Reconstruction. Prior feed-forward reconstruction frameworks typically generate 3D Gaussians for every pixel and are therefore constrained by both the number and the resolution of input views. For example, GS-LRM (Zhang et al., 2024e) operates on 2–4 images at 512×512 resolution, while AnySplat (Jiang et al., 2025a) is trained with 24 images at 448×448 resolution. By contrast, our video model synthesizes $V \times L = 6 \times 121 = 726$ input views at 704×1280 resolution, a scale far beyond the capacity of existing methods and beyond practical GPU memory limits. The bottleneck is the attention mechanism over visual tokens, whose memory and computation increase with the number of pixels. We therefore avoid scaling in pixel space and instead operate directly in the compressed video latent space produced by the camera-controlled video diffusion model. The resulting multi-view video latents are denoted $\mathbf{Z} \in \mathbb{R}^{V \times L' \times C \times h \times w}$.

Architecture. The 3DGS decoder $\mathcal{D}_s$ first projects its inputs into the hidden dimension of the main reconstruction blocks and ultimately predicts per-pixel 3D Gaussian features $\mathbf{G} \in \mathbb{R}^{V \times L \times H \times W \times 14}$. Its inputs are the multi-view video latents $\mathbf{Z}$ and encoded Plücker embeddings $\mathbf{E}$, both represented in latent dimensions $\mathbf{Z}, \mathbf{E} \in \mathbb{R}^{V \times L' \times C \times h \times w}$. Each input component is patchified to match the hidden dimension of the reconstruction blocks. No additional visual encoder is required; a spatial 2×2 patchification layer (Dosovitskiy et al., 2021) suffices to transform the video latents into flattened reconstruction tokens. The summed inputs are processed by reconstruction blocks following Long-LRM (Ziwen et al., 2024), where each block consists of one Transformer (Vaswani et al., 2017) layer followed by seven Mamba-2 (Dao & Gu, 2024) layers. Repeating this block twice yields 16 layers with 512 hidden dimensions. Finally, a transposed 3D convolution maps the hidden representation to 14 Gaussian channels: 3D position (x, y, z) , scale (s_x, s_y, s_z) , rotation quaternion (q_w, q_x, q_y, q_z) , opacity α , and RGB (r, g, b) . Formally, the static decoder is expressed as $\mathbf{G} = \mathcal{D}_s(\mathbf{Z}, \mathbf{E})$.

Plücker embeddings. Raw Plücker embeddings are first computed from camera poses $\{\mathbf{C}^{t,v}\}$ and intrinsics as $\mathbf{E}^{\text{raw}} \in \mathbb{R}^{V \times L \times 6 \times H \times W}$. We reuse the RGB encoder $\mathcal{E}$ from the pre-trained VAE to encode these Plücker embeddings. Specifically, the 3-dimensional ray directions and the 3-dimensional cross product of ray directions and origins are encoded separately with $\mathcal{E}$ and then concatenated in the latent channel dimension, producing $\mathbf{E}^{\text{enc}} \in \mathbb{R}^{V \times L' \times 2C \times h \times w}$. A lightweight MLP maps $\mathbf{E}^{\text{enc}}$ to $\mathbf{E} \in \mathbb{R}^{V \times L' \times C \times h \times w}$, reducing the channel dimension by a factor of 2 so that it matches the video latent $\mathbf{Z}$.

4.2 LOSS FUNCTION

Image-based supervision. We supervise the 3DGS decoder with an image-based reconstruction objective, decomposed into a Mean Squared Error (MSE) loss $\mathcal{L}_{mse}$ and an LPIPS loss $\mathcal{L}_{lips}$ (Zhang et al., 2018) computed with VGG (Simonyan & Zisserman, 2014) as the feature extractor.

Depth supervision. We observe that training the decoder with only RGB reconstruction losses often yields flattened geometry. To counteract this degeneracy, we additionally supervise rendered depth maps using temporally consistent video depth estimated by the off-the-shelf system ViPE (Huang et al., 2025), together with the scale-invariant depth loss $\mathcal{L}_{depth}$ from Long-LRM (Ziwen et al., 2024).

Opacity-based pruning. Following Long-LRM (Ziwen et al., 2024), we apply L1 regularization to opacity through $\mathcal{L}_{opacity}$ and remove the Gaussians with the lowest 80% opacity.

Total loss. The total objective is the weighted sum of all loss terms, with weights λ_i :

$$\mathcal{L} = \lambda_{mse} \mathcal{L}_{mse} + \lambda_{lips} \mathcal{L}_{lips} + \lambda_{depth} \mathcal{L}_{depth} + \lambda_{opacity} \mathcal{L}_{opacity} \quad (1)$$

where we set $\lambda_{mse} = 1.0$, $\lambda_{lips} = 0.5$, $\lambda_{depth} = 0.05$, and $\lambda_{opacity} = 0.1$.

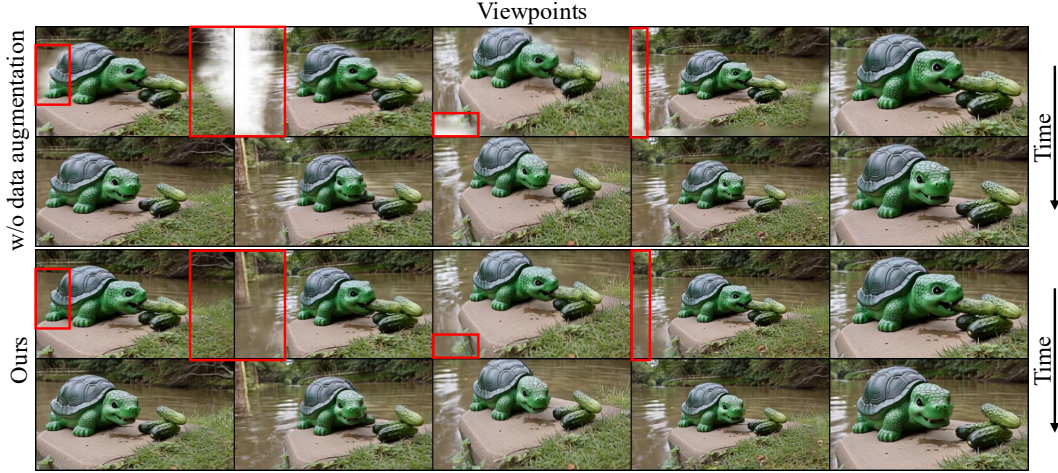


Figure 5: **Dynamic data augmentation.** Naively supervising the time-aware 3DGS decoder produces artifacts in the generated 3D Gaussians: early timesteps viewed from extreme poses exhibit low opacity in regions not covered by the available supervision. We therefore augment supervision with a motion-reversed video, ensuring that every timestep is observed with full spatial coverage and thereby preventing low-opacity artifacts at early timesteps.

5 EXTENSION TO DYNAMIC 3D SCENES

The same framework can be adapted to dynamic 3D scenes with only minimal modifications, which we describe below.

Self-distillation for dynamic scenes. The dynamic 3D setting closely mirrors the static 3D formulation. Rather than conditioning on a single image, the video model receives a single video of length L together with corresponding camera poses and generates multi-view video latents that preserve the same underlying motion. For video inputs, we follow a protocol analogous to the static case: diverse text prompts are sampled with large language models, videos are synthesized using video diffusion models (Cosmos, 2025; Wan, 2025), and the resulting videos are annotated with camera poses and depth maps using ViPE (Huang et al., 2025).

Dynamic 3DGS decoder architecture. Following the bullet-time design of Liang et al. (2025b), we define a dynamic 3DGS decoder $\mathcal{D}_s$ that produces time-dependent 3D Gaussians for all video frames. Architecturally, $\mathcal{D}_s$ largely follows $\mathcal{D}_s$, except that its input is augmented with encoded source and target time embeddings $\mathbf{T}^{\text{src}}, \mathbf{T}^{\text{tgt}} \in \mathbb{R}^{V \times L' \times C \times h \times w}$. The unencoded source times are assigned to each input frame, forming $\mathbf{T}^{\text{src, raw}} \in \mathbb{R}^{V \times L \times 1 \times H \times W}$, while the target time is represented as $\mathbf{T}^{\text{tgt, raw}} \in \mathbb{R}^{V \times 1 \times 1 \times H \times W}$. Both tensors are spatially repeated, augmented with a 2-dimensional sinusoidal embedding so that the channel dimension becomes 3, and encoded by the RGB encoder $\mathcal{E}$ into latent tensors $\mathbf{T}^{\text{src}}, \mathbf{T}^{\text{tgt}} \in \mathbb{R}^{V \times L' \times C \times h \times w}$. The target time is repeated along the latent temporal dimension to match L' . During training, we randomly sample a target timestep, decode its corresponding 3D Gaussians, and supervise them across all trajectories produced by dynamic data augmentation. We fine-tune $\mathcal{D}_d$ from a pre-trained $\mathcal{D}_s$, initializing the patchification layers for $\mathbf{T}^{\text{src}}$ and $\mathbf{T}^{\text{tgt}}$ to zero. The tensors $\mathbf{T}^{\text{src}}$ and $\mathbf{T}^{\text{tgt}}$ are directly added to the sum of $\mathbf{Z}$ and $\mathbf{E}$. Formally, the dynamic decoder is defined as $\mathbf{G} = \mathcal{D}_d(\mathbf{Z}, \mathbf{E}, \mathbf{T}^{\text{src}}, \mathbf{T}^{\text{tgt}})$.

Dynamic data augmentation. In static 3D scenes, where time is irrelevant, frames from different timesteps can all supervise the same 3DGS rendering. For dynamic 3D scenes, by contrast, the 3DGS varies over time, so only frames at the corresponding timestamp provide valid supervision. This constraint can induce a trivial solution in which the 3DGS for a given timestep ignores information from other timesteps, as illustrated in Fig. 5.

To address this imbalance, we introduce a dynamic data augmentation strategy that distributes supervision more evenly across timesteps. Early frames are naturally associated with viewpoints close to the input image, whereas later frames correspond to more distant viewpoints; we therefore augment training with paired supervision views for each timestep, one near and one far. Concretely, we reverse the input video in temporal order and feed it into the video model $\mathcal{V}$, producing six

Table 1: **State-of-the-art comparisons.** We compare our method with prior single-image-to-3D generation approaches on RealEstate10K, DL3DV, and Tanks-and-Temples.

Method	RealEstate10K			DL3DV			Tanks-and-Temples		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
ZeroNVS	13.01	0.378	0.448	13.35	0.339	0.465	12.94	0.325	0.470
ViewCrafter	16.84	0.514	0.341	15.53	0.525	0.352	14.93	0.483	0.384
Wonderland	17.15	0.550	0.292	16.64	0.574	0.325	15.90	0.510	0.344
Bolt3D	21.54	0.747	0.234	-	-	-	-	-	-
Ours	21.79	0.752	0.219	20.09	0.583	0.313	19.24	0.570	0.336



Figure 6: **Image-to-3DGS Generation.** We visualize five rendered viewpoints from generated 3DGS scenes, illustrating the method’s novel-view synthesis behavior.

additional multi-view sequences. After reversing these sequences back into the original motion order, we obtain six trajectories that progress inward, from far viewpoints toward the input. Together with the original six outward trajectories, this procedure yields 12 supervision views per timestep. This augmentation is used only during training: the reversed trajectories act solely as additional supervision signals that prevent collapse caused by pruning artifacts and are not required at inference time.

6 EXPERIMENTS

6.1 EXPERIMENTAL SETUP

Datasets. For training, we deliberately avoid existing multi-view datasets and instead construct a dedicated *Lyra* dataset in which supervision for the 3DGS decoder is supplied entirely by the video model. The 3D reconstruction setting contains 59,031 images, while the 4D setting contains 7,378 videos. All examples are derived from diverse text prompts spanning indoor and outdoor environments, humans, animals, and both realistic and imaginative content. For each image in the 3D setting, or each video in the 4D setting, we synthesize six camera trajectories, producing 354,186 videos for 3D training and 44,268 videos for 4D training.

Baselines. We compare against state-of-the-art methods for generative single-image-to-3D reconstruction, namely ZeroNVS (Sargent et al., 2024), ViewCrafter (Yu et al., 2024b), Wonderland (Liang et al., 2025a), and Bolt3D (Szymanowicz et al., 2025b). Because *no source code is available* for evaluating these methods on our out-of-distribution evaluation set, our quantitative comparison primarily relies on the numbers reported in the respective papers.

Evaluation. Following prior work, we evaluate single-image-to-3D generation on RealEstate10K (Zhou et al., 2018), DL3DV (Ling et al., 2024b), and Tanks and Temples (Knapitsch et al., 2017). We adopt the evaluation protocol used by Wonderland (Liang et al., 2025a) and Bolt3D (Szymanowicz et al., 2025b), and report standard reconstruction metrics: PSNR, SSIM, and LPIPS.



Figure 7: **Ablations.** We visualize each ablation by rendering the same extreme novel viewpoint after image-to-3DGS generation; additional depth visualizations and ablation evidence are provided in Appendix C.4.

6.2 MAIN RESULTS

Quantitative results. Tab. 1 reports quantitative results for single-image-to-3D generation. Across all benchmarks and all metrics, our method outperforms the prior approaches under comparison. These results indicate that further improvements in reconstruction quality are likely to follow from stronger video generative models, because our reconstructions directly inherit and operationalize the supervision they provide. Additional quantitative comparisons on the *Lyra* dataset are included in Appendix C.

Qualitative results. We show novel-view renderings from generated 3DGS scenes in Fig. 1 and Fig. 6; video results will be released upon acceptance. The generated 3D/4D scenes exhibit high-quality novel-view content in previously unseen regions while remaining consistent with the input image or video. Qualitative comparisons on the *Lyra* dataset are provided in Appendix C.

6.3 ABLATIONS

Tab. 2 ablates key design choices on diverse out-of-distribution prompts. In each case, we compare 3DGS renderings against videos generated by the camera-controlled video diffusion model that supplies supervision.

Real data only. Instead of self-distillation, this variant trains only on real multi-view datasets, namely RealEstate10K (Zhou et al., 2018) and DL3DV (Ling et al., 2024b). Although these datasets are widely used in prior work, they yield limited generalization to out-of-distribution scenes.

Self-distillation + real data. Jointly training with self-distillation and real data, perhaps surprisingly, does not improve over self-distillation alone. This outcome suggests that the synthetic supervision produced by self-distillation is sufficiently diverse and consistent to train the reconstruction model without additional real multi-view data.

No depth loss. Depth supervision prevents degenerate flat geometry and also yields a modest improvement in image-based metrics. Depth-map visualizations are provided in Appendix C.4.

No opacity pruning. This variant retains all Gaussians rather than pruning them according to opacity. Learning to remove most low-opacity 3D Gaussians makes the representation more compact and slightly improves visual quality: rendering at $H = 704$ and $W = 1280$ requires 18ms with pruning compared with 30ms without pruning, corresponding to a $1.67\times$ speedup.

No LPIPS loss. Including the LPIPS loss improves robustness to input inconsistencies and better preserves high-frequency details.

No multi-view fusion. A naive alternative is to predict 3D Gaussians independently for each camera trajectory and then fuse those Gaussians into a single point cloud. In practice, however, we observe

Table 2: **Ablation study on the *Lyra* dataset.**

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Ours	24.77	0.837	0.224
Data			
real data only	19.08	0.659	0.413
self-distill. + real data	24.74	0.823	0.236
Loss			
w/o depth loss	24.31	0.811	0.247
w/o opacity pruning	24.55	0.820	0.237
w/o LPIPS loss	23.74	0.766	0.370
Architecture			
w/o multi-view fusion	17.73	0.632	0.446
w/o Mamba-2	24.58	0.818	0.241
w/o latent 3DGS		OOM	

substantial benefits from learning multi-view fusion across trajectories within the reconstruction blocks, where tokens can attend to one another.

No Mamba-2. Replacing the joint Transformer/Mamba-2 design with Transformer-only blocks produces slightly lower visual quality. For a single forward pass with $V = 6$, $L = 121$, $H = 704$, and $W = 1280$, the joint blocks require 3213ms, whereas Transformer-only blocks require 20922ms, giving a $6.5 \times$ speedup.

No latent-based 3DGS. Pixel-space methods such as BTimer (Liang et al., 2025b) are restricted to 12 input frames, whereas our latent-space decoder can process up to 726 input frames. Consequently, moving the 3DGS decoder from latent space to pixel space results in out-of-memory failure.

7 CONCLUSION

In this work, we introduced *Lyra*, a 3D and 4D generation framework that requires only a single image or video as input. Rather than collecting multi-view datasets, we use camera-controlled video diffusion models as teachers in a self-distillation framework for a student 3DGS decoder. Because the decoder operates in the latent space of the video model, it can directly reconstruct 3D Gaussians without post-processing or per-scene optimization. This design improves generalizability and scene coverage across diverse inputs, while the dynamic extension demonstrates the feasibility of 4D generation from monocular video. At present, the spatial scale and consistency of the generated scenes remain bounded by the capacity of the camera-controlled video diffusion model. Accordingly, improvements to the underlying video diffusion model should translate into more consistent large-scale scene synthesis. Exploring the incorporation of auto-regressive techniques (Chen et al., 2024a) into our framework is another promising direction for large-scale generation. Finally, explicitly modeling motion and tracking information within the reconstruction network, as in concurrent work (Lin et al., 2025), may further improve visual motion quality.

8 ETHICS STATEMENT

As with generative AI technologies more broadly, our method carries potential misuse risks, including the production of misleading or synthetic 3D/4D content. Nevertheless, the primary objective of this research is to advance simulation, robotics, and embodied AI by providing scalable and controllable tools for data generation. We therefore expect the most consequential contributions of this work to arise from the following capabilities and insights.

- **The ability to enable realistic simulation for embodied agents:** We provide controllable and physically consistent camera motion within generated 3D and 4D environments, thereby supporting the training and evaluation of agents that must perceive, navigate, and interact with complex scenes.
- **Improved scalability for data engines:** Our method removes the need for multi-view capture and expensive per-scene optimization, while facilitating the large-scale production of high-fidelity and customizable scenes for robotics, reinforcement learning, and closed-loop simulation research.
- **Better technical understanding:** As an academic study, this work offers a foundational case study of how camera control interacts with spatiotemporal generative models, with implications for research in computer vision, graphics, and embodied AI.

Although we caution against deceptive or otherwise wrongful uses of this technology, especially those in which realistic generative outputs could produce misleading or unverifiable media, we believe that appropriate safeguards, including provenance tracking, dataset documentation, and rigorous evaluation practices, can help ensure that generative models advance science and society in beneficial ways.

9 REPRODUCIBILITY STATEMENT

To enable full reproduction of our results and to accelerate subsequent research in this area, we will release the training and inference code, model weights, and data for both 3D and 4D generation upon acceptance.

REFERENCES

- Sherwin Bahmani, Jeong Joon Park, Despoina Paschalidou, Hao Tang, Gordon Wetzstein, Leonidas Guibas, Luc Van Gool, and Radu Timofte. 3d-aware video generation. In *TMLR*, 2023a.
- Sherwin Bahmani, Jeong Joon Park, Despoina Paschalidou, Xingguang Yan, Gordon Wetzstein, Leonidas Guibas, and Andrea Tagliasacchi. CC3D: Layout-conditioned generation of compositional 3D scenes. In *Proc. ICCV*, 2023b.
- Sherwin Bahmani, Xian Liu, Wang Yifan, Ivan Skorokhodov, Victor Rong, Ziwei Liu, Xihui Liu, Jeong Joon Park, Sergey Tulyakov, Gordon Wetzstein, Andrea Tagliasacchi, and David B. Lindell. Tc4d: Trajectory-conditioned text-to-4d generation. In *Proc. ECCV*, 2024a.
- Sherwin Bahmani, Ivan Skorokhodov, Victor Rong, Gordon Wetzstein, Leonidas Guibas, Peter Wonka, Sergey Tulyakov, Jeong Joon Park, Andrea Tagliasacchi, and David B. Lindell. 4d-fy: Text-to-4d generation using hybrid score distillation sampling. In *Proc. CVPR*, 2024b.
- Sherwin Bahmani, Ivan Skorokhodov, Guocheng Qian, Aliaksandr Siarohin, Willi Menapace, Andrea Tagliasacchi, David B Lindell, and Sergey Tulyakov. Ac3d: Analyzing and improving 3d camera control in video diffusion transformers. *Proc. CVPR*, 2025a.
- Sherwin Bahmani, Ivan Skorokhodov, Aliaksandr Siarohin, Willi Menapace, Guocheng Qian, Michael Vasilkovsky, Hsin-Ying Lee, Chaoyang Wang, Jiaxu Zou, Andrea Tagliasacchi, et al. Vd3d: Taming large video diffusion transformers for 3d camera control. *Proc. ICLR*, 2025b.
- Jianhong Bai, Menghan Xia, Xiao Fu, Xintao Wang, Lianrui Mu, Jinwen Cao, Zuozhu Liu, Haoji Hu, Xiang Bai, Pengfei Wan, et al. Recammaster: Camera-controlled generative rendering from a single video. *arXiv preprint arXiv:2503.11647*, 2025a.
- Jianhong Bai, Menghan Xia, Xintao Wang, Ziyang Yuan, Xiao Fu, Zuozhu Liu, Haoji Hu, Pengfei Wan, and Di Zhang. Syncammaster: Synchronizing multi-camera video generation from diverse viewpoints. *Proc. ICLR*, 2025b.
- Jinze Bai, Shuai Bai, Shusheng Yang, Shijie Wang, Sinan Tan, Peng Wang, Junyang Lin, Chang Zhou, and Jingren Zhou. Qwen-vl: A versatile vision-language model for understanding, localization, text reading, and beyond. *arXiv preprint arXiv:2308.12966*, 2023.
- Black Forest Labs. Flux. <https://github.com/black-forest-labs/flux>, 2024.
- Andreas Blattmann, Tim Dockhorn, Sumith Kulal, Daniel Mendelevitch, Maciej Kilian, Dominik Lorenz, Yam Levi, Zion English, Vikram Voleti, Adam Letts, et al. Stable video diffusion: Scaling latent video diffusion models to large datasets. *arXiv preprint arXiv:2311.15127*, 2023.
- Chenjie Cao, Jingkai Zhou, Shikai Li, Jingyun Liang, Chaohui Yu, Fan Wang, Xiangyang Xue, and Yanwei Fu. Uni3c: Unifying precisely 3d-enhanced camera and human motion controls for video generation. *arXiv preprint arXiv:2504.14899*, 2025.
- Yukang Cao, Liang Pan, Kai Han, Kwan-Yee K Wong, and Ziwei Liu. Avatargo: Zero-shot 4d human-object interaction generation and animation. *arXiv preprint arXiv:2410.07164*, 2024.
- Zenghao Chai, Chen Tang, Yongkang Wong, and Mohan Kankanhalli. Star: Skeleton-aware text-based 4d avatar generation with in-network motion retargeting. *arXiv preprint arXiv:2406.04629*, 2024.
- Eric R Chan, Connor Z Lin, Matthew A Chan, Koki Nagano, Boxiao Pan, Shalini De Mello, Orazio Gallo, Leonidas J Guibas, Jonathan Tremblay, Sameh Khamis, et al. Efficient geometry-aware 3D generative adversarial networks. In *Proc. CVPR*, 2022.
- Eric R Chan, Koki Nagano, Matthew A Chan, Alexander W Bergman, Jeong Joon Park, Axel Levy, Miika Aittala, Shalini De Mello, Tero Karras, and Gordon Wetzstein. Generative novel view synthesis with 3d-aware diffusion models. In *Proc. ICCV*, 2023.
- David Charatan, Sizhe Lester Li, Andrea Tagliasacchi, and Vincent Sitzmann. pixelsplat: 3d gaussian splats from image pairs for scalable generalizable 3d reconstruction. In *Proc. CVPR*, 2024.

-
- Boyuan Chen, Diego Martí Monsó, Yilun Du, Max Simchowitz, Russ Tedrake, and Vincent Sitzmann. Diffusion forcing: Next-token prediction meets full-sequence diffusion. *Proc. NeurIPS*, 2024a.
- Ce Chen, Shaoli Huang, Xuelin Chen, Guangyi Chen, Xiaoguang Han, Kun Zhang, and Mingming Gong. Ct4d: Consistent text-to-4d generation with animatable meshes. *arXiv preprint arXiv:2408.08342*, 2024b.
- Haoxin Chen, Menghan Xia, Yingqing He, Yong Zhang, Xiaodong Cun, Shaoshu Yang, Jinbo Xing, Yaofang Liu, Qifeng Chen, Xintao Wang, Chao Weng, and Ying Shan. Videocrafter1: Open diffusion models for high-quality video generation. *arXiv preprint arXiv:2310.19512*, 2023a.
- Kevin Chen, Christopher B Choy, Manolis Savva, Angel X Chang, Thomas Funkhouser, and Silvio Savarese. Text2Shape: Generating shapes from natural language by learning joint embeddings. In *Proc. ACCV*, 2018.
- Rui Chen, Yongwei Chen, Ningxin Jiao, and Kui Jia. Fantasia3D: Disentangling geometry and appearance for high-quality text-to-3D content creation. *arXiv preprint arXiv:2303.13873*, 2023b.
- Soon Yau Cheong, Duygu Ceylan, Armin Mustafa, Andrew Gilbert, and Chun-Hao Paul Huang. Boosting camera motion control for video diffusion transformers. *arXiv preprint arXiv:2410.10802*, 2024.
- Wen-Hsuan Chu, Lei Ke, and Katerina Fragkiadaki. Dreamscene4d: Dynamic multi-object scene generation from monocular videos. *arXiv preprint arXiv:2405.02280*, 2024.
- Team Cosmos. Cosmos world foundation model platform for physical ai. *arXiv preprint arXiv:2501.03575*, 2025.
- Tri Dao and Albert Gu. Transformers are ssms: Generalized models and efficient algorithms through structured state space duality. *arXiv preprint arXiv:2405.21060*, 2024.
- Terrance DeVries, Miguel Angel Bautista, Nitish Srivastava, Graham W Taylor, and Joshua M Susskind. Unconstrained scene generation with locally conditioned radiance fields. In *Proc. ICCV*, 2021.
- Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. An image is worth 16x16 words: Transformers for image recognition at scale. In *Proc. ICLR*, 2021.
- Qijun Feng, Zhen Xing, Zuxuan Wu, and Yu-Gang Jiang. FDGaussian: Fast Gaussian splatting from single image via geometric-aware diffusion model. *arXiv preprint arXiv:2403.10242*, 2024a.
- Yutao Feng, Yintong Shang, Xiang Feng, Lei Lan, Shandian Zhe, Tianjia Shao, Hongzhi Wu, Kun Zhou, Hao Su, Chenfanfu Jiang, et al. Elastogen: 4d generative elastodynamics. *arXiv preprint arXiv:2405.15056*, 2024b.
- Quankai Gao, Qiangeng Xu, Zhe Cao, Ben Mildenhall, Wenchao Ma, Le Chen, Danhang Tang, and Ulrich Neumann. Gaussianflow: Splatting Gaussian dynamics for 4D content creation. *arXiv preprint arXiv:2403.12365*, 2024a.
- Ruiqi Gao, Aleksander Holynski, Philipp Henzler, Arthur Brussee, Ricardo Martin-Brualla, Pratul Srinivasan, Jonathan T Barron, and Ben Poole. Cat3d: Create anything in 3d with multi-view diffusion models. In *Proc. NeurIPS*, 2024b.
- William Gao, Noam Aigerman, Thibault Groueix, Vova Kim, and Rana Hanocka. TextDeformer: Geometry manipulation using text guidance. In *SIGGRAPH*, 2023.
- Jiatao Gu, Alex Trevithick, Kai-En Lin, Joshua M Susskind, Christian Theobalt, Lingjie Liu, and Ravi Ramamoorthi. NerfDiff: Single-image view synthesis with NeRF-guided distillation from 3D-aware diffusion. In *Proc. ICML*, 2023.
- Zekai Gu, Rui Yan, Jiahao Lu, Peng Li, Zhiyang Dou, Chenyang Si, Zhen Dong, Qifeng Liu, Cheng Lin, Ziwei Liu, et al. Diffusion as shader: 3d-aware video diffusion for versatile video generation control. In *SIGGRAPH*, 2025.

Junlin Han, Filippos Kokkinos, and Philip Torr. VFusion3D: Learning scalable 3D generative models from video diffusion models. *arXiv preprint arXiv:2403.12034*, 2024.

Hao He, Yinghao Xu, Yuwei Guo, Gordon Wetzstein, Bo Dai, Hongsheng Li, and Ceyuan Yang. Cameractrl: Enabling camera control for text-to-video generation. *arXiv preprint arXiv:2404.02101*, 2024a.

Hao He, Ceyuan Yang, Shanchuan Lin, Yinghao Xu, Meng Wei, Liangke Gui, Qi Zhao, Gordon Wetzstein, Lu Jiang, and Hongsheng Li. Cameractrl ii: Dynamic scene exploration via camera-controlled video diffusion models. *arXiv preprint arXiv:2503.10592*, 2025.

Xianglong He, Junyi Chen, Sida Peng, Di Huang, Yangguang Li, Xiaoshui Huang, Chun Yuan, Wanli Ouyang, and Tong He. GVGEn: Text-to-3D generation with volumetric representation. *arXiv preprint arXiv:2403.12957*, 2024b.

Lukas Höllein, Ang Cao, Andrew Owens, Justin Johnson, and Matthias Nießner. Text2room: Extracting textured 3d meshes from 2d text-to-image models. In *Proc. ICCV*, 2023.

Lukas Höllein, Aljaž Božič, Norman Müller, David Novotny, Hung-Yu Tseng, Christian Richardt, Michael Zollhöfer, and Matthias Nießner. ViewDiff: 3D-consistent image generation with text-to-image models. In *Proc. CVPR*, 2024.

Yicong Hong, Kai Zhang, Jiuxiang Gu, Sai Bi, Yang Zhou, Difan Liu, Feng Liu, Kalyan Sunkavalli, Trung Bui, and Hao Tan. LRM: Large reconstruction model for single image to 3D. In *Proc. ICLR*, 2024.

Chen Hou, Guoqiang Wei, Yan Zeng, and Zhibo Chen. Training-free camera control for video generation. *arXiv preprint arXiv:2406.10126*, 2024.

Ronghang Hu, Nikhila Ravi, Alexander C Berg, and Deepak Pathak. Worldsheet: Wrapping the world in a 3d sheet for view synthesis from a single image. In *Proc. ICCV*, 2021.

Tao Hu, Haoyang Peng, Xiao Liu, and Yuewen Ma. Ex-4d: Extreme viewpoint 4d video synthesis via depth watertight mesh. *arXiv preprint arXiv:2506.05554*, 2025.

Teng Hu, Jiangning Zhang, Ran Yi, Yating Wang, Hongrui Huang, Jieyu Weng, Yabiao Wang, and Lizhuang Ma. Motionmaster: Training-free camera motion transfer for video generation. *arXiv preprint arXiv:2404.15789*, 2024.

Chun-Hao Paul Huang, Jae Shin Yoon, Hyeonho Jeong, Niloy Mitra, and Duygu Ceylan. Camera pose estimation emerging in video diffusion transformer, 2024a. URL <https://openreview.net/forum?id=lgf2LW7fOJ>.

Jiahui Huang, Qunjie Zhou, Hesam Rabeti, Aleksandr Korovko, Huan Ling, Xuanchi Ren, Tianchang Shen, Jun Gao, Dmitry Slepichev, Chen-Hsuan Lin, Jiawei Ren, Kevin Xie, Joydeep Biswas, Laura Leal-Taixe, and Sanja Fidler. Vipe: Video pose engine for 3d geometric perception. In *NVIDIA Research Whitepapers*, 2025.

Tianyu Huang, Yihan Zeng, Hui Li, Wangmeng Zuo, and Rynson WH Lau. Dreamphysics: Learning physical properties of dynamic 3d gaussians with video diffusion priors. *arXiv preprint arXiv:2406.01476*, 2024b.

Ajay Jain, Ben Mildenhall, Jonathan T Barron, Pieter Abbeel, and Ben Poole. Zero-shot text-guided object generation with dream fields. In *Proc. CVPR*, 2022.

Nikolay Jetchev. ClipMatrix: Text-controlled creation of 3D textured meshes. *arXiv preprint arXiv:2109.12922*, 2021.

Lihan Jiang, Yucheng Mao, Linning Xu, Tao Lu, Kerui Ren, Yichen Jin, Xudong Xu, Mulin Yu, Jiangmiao Pang, Feng Zhao, et al. Anysplat: Feed-forward 3d gaussian splatting from unconstrained views. *arXiv preprint arXiv:2505.23716*, 2025a.

Lutao Jiang and Lin Wang. Brightdreamer: Generic 3D Gaussian generative framework for fast text-to-3D synthesis. *arXiv preprint arXiv:2403.11273*, 2024.

-
- Yanqin Jiang, Chaohui Yu, Chenjie Cao, Fan Wang, Weiming Hu, and Jin Gao. Animate3d: Animating any 3d model with multi-view video diffusion. *arXiv preprint arXiv:2407.11398*, 2024a.
- Yanqin Jiang, Li Zhang, Jin Gao, Weimin Hu, and Yao Yao. Consistent4d: Consistent 360 $\{\backslash\deg\}$ dynamic object generation from monocular video. In *Proc. ICLR*, 2024b.
- Zeren Jiang, Chuanxia Zheng, Iro Laina, Diane Larlus, and Andrea Vedaldi. Geo4d: Leveraging video generators for geometric 4d scene reconstruction. *arXiv preprint arXiv:2504.07961*, 2025b.
- Wonjoon Jin, Qi Dai, Chong Luo, Seung-Hwan Baek, and Sunghyun Cho. Flovd: Optical flow meets video diffusion model for enhanced camera-controlled video synthesis. In *Proc. CVPR*, 2025.
- Xuan Ju, Weicai Ye, Quande Liu, Qiulin Wang, Xintao Wang, Pengfei Wan, Di Zhang, Kun Gai, and Qiang Xu. Fulldit: Multi-task video generative foundation model with full attention. *arXiv preprint arXiv:2503.19907*, 2025.
- Yash Kant, Aliaksandr Siarohin, Ziyi Wu, Michael Vasilkovsky, Guocheng Qian, Jian Ren, Riza Alp Guler, Bernard Ghanem, Sergey Tulyakov, and Igor Gilitschenski. Spad: Spatially aware multi-view diffusers. In *Proc. CVPR*, 2024.
- Yash Kant, Ethan Weber, Jin Kyu Kim, Rawal Khirodkar, Su Zhaoen, Julieta Martinez, Igor Gilitschenski, Shunsuke Saito, and Timur Bagautdinov. Pippo: High-resolution multi-view humans from a single image. In *Proc. CVPR*, 2025.
- Oren Katzir, Or Patashnik, Daniel Cohen-Or, and Dani Lischinski. Noise-free score distillation. In *Proc. ICLR*, 2024.
- Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 3d gaussian splatting for real-time radiance field rendering. In *ACM TOG*, 2023.
- Seung Wook Kim, Bradley Brown, Kangxue Yin, Karsten Kreis, Katja Schwarz, Daiqing Li, Robin Rombach, Antonio Torralba, and Sanja Fidler. NeuralField-LDM: Scene generation with hierarchical latent diffusion models. In *Proc. CVPR*, 2023.
- Diederik P Kingma and Max Welling. Auto-encoding variational bayes. *arXiv preprint arXiv:1312.6114*, 2013.
- Mert Kiray, Paul Uhlenbruck, Nassir Navab, and Benjamin Busam. Promptvfx: Text-driven fields for open-world 3d gaussian animation. *arXiv preprint arXiv:2506.01091*, 2025.
- Tobias Kirschstein, Javier Romero, Artem Sevastopolsky, Matthias Nießner, and Shunsuke Saito. Avat3r: Large animatable gaussian reconstruction model for high-fidelity 3d head avatars. *arXiv preprint arXiv:2502.20220*, 2025.
- Arno Knapitsch, Jaesik Park, Qian-Yi Zhou, and Vladlen Koltun. Tanks and temples: Benchmarking large-scale scene reconstruction. *ACM TOG*, 2017.
- Zhengfei Kuang, Shengqu Cai, Hao He, Yinghao Xu, Hongsheng Li, Leonidas Guibas, and Gordon Wetzstein. Collaborative video diffusion: Consistent multi-video generation with camera control. In *Proc. NeurIPS*, 2024.
- Kyungmin Lee, Kihyuk Sohn, and Jinwoo Shin. DreamFlow: High-quality text-to-3D generation by approximating probability flow. In *Proc. ICLR*, 2024a.
- Yao-Chih Lee, Yi-Ting Chen, Andrew Wang, Ting-Hsuan Liao, Brandon Y Feng, and Jia-Bin Huang. Vividdream: Generating 3d scene with ambient dynamics. *arXiv preprint arXiv:2405.20334*, 2024b.
- Guojun Lei, Chi Wang, Hong Li, Rong Zhang, Yikai Wang, and Weiwei Xu. Animateanything: Consistent and controllable animation for video generation. *arXiv preprint arXiv:2411.10836*, 2024.
- Guojun Lei, Chi Wang, Rong Zhang, Yikai Wang, Hong Li, and Weiwei Xu. Animateanything: Consistent and controllable animation for video generation. In *Proc. CVPR*, 2025.

- Bing Li, Cheng Zheng, Wenxuan Zhu, Jinjie Mai, Biao Zhang, Peter Wonka, and Bernard Ghanem. Vivid-zoo: Multi-view video generation with diffusion model. *arXiv preprint arXiv:2406.08659*, 2024a.
- Jiahao Li, Hao Tan, Kai Zhang, Zexiang Xu, Fujun Luan, Yinghao Xu, Yicong Hong, Kalyan Sunkavalli, Greg Shakhnarovich, and Sai Bi. Instant3D: Fast text-to-3D with sparse-view generation and large reconstruction model. In *Proc. ICLR*, 2024b.
- Jinwei Li, Huan-ang Gao, Wenyi Li, Haohan Chi, Chenyu Liu, Chenxi Du, Yiqian Liu, Mingju Gao, Guiyu Zhang, Zongzheng Zhang, et al. Fb-4d: Spatial-temporal coherent dynamic 3d content generation with feature banks. *arXiv preprint arXiv:2503.20784*, 2025a.
- Longfei Li, Zhiwen Fan, Wenyan Cong, Xinhang Liu, Yuyang Yin, Matt Foutter, Panwang Pan, Chenyu You, Yue Wang, Zhangyang Wang, et al. Martian world models: Controllable video synthesis with physically accurate 3d reconstructions. *arXiv preprint arXiv:2507.07978*, 2025b.
- Renjie Li, Panwang Pan, Bangbang Yang, Dejia Xu, Shijie Zhou, Xuanyang Zhang, Zeming Li, Achuta Kadambi, Zhangyang Wang, and Zhiwen Fan. 4k4dgen: Panoramic 4d generation at 4k resolution. *arXiv preprint arXiv:2406.13527*, 2024c.
- Teng Li, Guangcong Zheng, Rui Jiang, Shuigen Zhan, Tao Wu, Yehao Lu, Yining Lin, Chuanyun Deng, Yapan Xiong, Min Chen, et al. Realcam-i2v: Real-world image-to-video generation with interactive complex camera control. *arXiv preprint arXiv:2502.10059*, 2025c.
- Tianye Li, Mira Slavcheva, Michael Zollhoefer, Simon Green, Christoph Lassner, Changil Kim, Tanner Schmidt, Steven Lovegrove, Michael Goesele, Richard Newcombe, et al. Neural 3d video synthesis from multi-view video. In *Proc. CVPR*, 2022.
- Zhiqi Li, Yiming Chen, and Peidong Liu. Dreammesh4d: Video-to-4d generation with sparse-controlled gaussian-mesh hybrid representation. *arXiv preprint arXiv:2410.06756*, 2024d.
- Zhiqi Li, Yiming Chen, Lingzhe Zhao, and Peidong Liu. Controllable text-to-3D generation via surface-aligned Gaussian splatting. *arXiv preprint arXiv:2403.09981*, 2024e.
- Hanwen Liang, Yuyang Yin, Dejia Xu, Hanxue Liang, Zhangyang Wang, Konstantinos N Plataniotis, Yao Zhao, and Yunchao Wei. Diffusion4d: Fast spatial-temporal consistent 4d generation via video diffusion models. *arXiv preprint arXiv:2405.16645*, 2024.
- Hanwen Liang, Junli Cao, Vidit Goel, Guocheng Qian, Sergei Korolev, Demetri Terzopoulos, Konstantinos N Plataniotis, Sergey Tulyakov, and Jian Ren. Wonderland: Navigating 3d scenes from a single image. *Proc. CVPR*, 2025a.
- Hanxue Liang, Jiawei Ren, Ashkan Mirzaei, Antonio Torralba, Ziwei Liu, Igor Gilitschenski, Sanja Fidler, Cengiz Oztireli, Huan Ling, Zan Gojcic, and Jiahui Huang. Feed-forward bullet-time reconstruction of dynamic scenes from monocular videos. *Proc. NeurIPS*, 2025b.
- Yixun Liang, Xin Yang, Jiantao Lin, Haodong Li, Xiaogang Xu, and Yingcong Chen. Lucid-dreamer: Towards high-fidelity text-to-3D generation via interval score matching. *arXiv preprint arXiv:2311.11284*, 2023.
- Chen-Hsuan Lin, Jun Gao, Luming Tang, Towaki Takikawa, Xiaohui Zeng, Xun Huang, Karsten Kreis, Sanja Fidler, Ming-Yu Liu, and Tsung-Yi Lin. Magic3D: High-resolution text-to-3D content creation. In *Proc. CVPR*, 2023a.
- Chenguo Lin, Yuchen Lin, Panwang Pan, Yifan Yu, Honglei Yan, Katerina Fragkiadaki, and Yadong Mu. Movies: Motion-aware 4d dynamic view synthesis in one second. *arXiv preprint arXiv:2507.10065*, 2025.
- Jiajing Lin, Zhenzhong Wang, Yongjie Hou, Yuzhou Tang, and Min Jiang. Phy124: Fast physics-driven 4d content generation from a single image. *arXiv preprint arXiv:2409.07179*, 2024.
- Yukang Lin, Haonan Han, Chaoqun Gong, Zunnan Xu, Yachao Zhang, and Xiu Li. Consistent123: One image to highly consistent 3D asset using case-aware diffusion priors. In *arXiv preprint arXiv:2309.17261*, 2023b.

- Huan Ling, Seung Wook Kim, Antonio Torralba, Sanja Fidler, and Karsten Kreis. Align your gaussians: Text-to-4d with dynamic 3d gaussians and composed diffusion models. In *Proc. CVPR*, 2024a.
- Lu Ling, Yichen Sheng, Zhi Tu, Wentian Zhao, Cheng Xin, Kun Wan, Lantao Yu, Qianyu Guo, Zixun Yu, Yawen Lu, et al. D13dv-10k: A large-scale scene dataset for deep learning-based 3d vision. In *Proc. CVPR*, 2024b.
- Pengyang Ling, Jiazi Bu, Pan Zhang, Xiaoyi Dong, Yuhang Zang, Tong Wu, Huaian Chen, Jiaqi Wang, and Yi Jin. Motionclone: Training-free motion cloning for controllable video generation. *arXiv preprint arXiv:2406.05338*, 2024c.
- Lijuan Liu, Wenfa Li, Dongbo Zhang, Shuo Wang, and Shaohui Jiao. Idcnet: Guided video diffusion for metric-consistent rgbd scene generation with precise camera control. *arXiv preprint arXiv:2508.04147*, 2025.
- Pengkun Liu, Yikai Wang, Fuchun Sun, Jiafang Li, Hang Xiao, Hongxiang Xue, and Xinzhou Wang. Isotropic3D: Image-to-3D generation based on a single clip embedding. *arXiv preprint arXiv:2403.10395*, 2024a.
- Ruoshi Liu, Rundi Wu, Basile Van Hoorick, Pavel Tokmakov, Sergey Zakharov, and Carl Vondrick. Zero-1-to-3: Zero-shot one image to 3D object. In *Proc. ICCV*, 2023.
- Xian Liu, Xiaohang Zhan, Jiayang Tang, Ying Shan, Gang Zeng, Dahua Lin, Xihui Liu, and Ziwei Liu. HumanGaussian: Text-driven 3D human generation with Gaussian splatting. In *Proc. CVPR*, 2024b.
- Yuan Liu, Cheng Lin, Zijiao Zeng, Xiaoxiao Long, Lingjie Liu, Taku Komura, and Wenping Wang. SyncDreamer: Generating multiview-consistent images from a single-view image. In *Proc. ICLR*, 2024c.
- Xiaoxiao Long, Yuan-Chen Guo, Cheng Lin, Yuan Liu, Zhiyang Dou, Lingjie Liu, Yuexin Ma, Song-Hai Zhang, Marc Habermann, Christian Theobalt, et al. Wonder3D: Single image to 3D using cross-domain diffusion. In *Proc. CVPR*, 2024.
- Yifan Lu, Xuanchi Ren, Jiawei Yang, Tianchang Shen, Zhangjie Wu, Jun Gao, Yue Wang, Siheng Chen, Mike Chen, Sanja Fidler, et al. Infinicube: Unbounded and controllable dynamic 3d driving scene generation with world-guided video models. *arXiv preprint arXiv:2412.03934*, 2024.
- Yue Ma, Kunyu Feng, Xinhua Zhang, Hongyu Liu, David Junhao Zhang, Jinbo Xing, Yinhan Zhang, Ayden Yang, Zeyu Wang, and Qifeng Chen. Follow-your-creation: Empowering 4d creation through video inpainting. *arXiv preprint arXiv:2506.04590*, 2025.
- Jinjie Mai, Wenxuan Zhu, Haozhe Liu, Bing Li, Cheng Zheng, Jürgen Schmidhuber, and Bernard Ghanem. Can video diffusion model reconstruct 4d geometry? *arXiv preprint arXiv:2503.21082*, 2025.
- Qiaowei Miao, Yawei Luo, and Yi Yang. Pla4d: Pixel-level alignments for text-to-4d gaussian splatting. *arXiv preprint arXiv:2405.19957*, 2024.
- Ben Mildenhall, Pratul P Srinivasan, Matthew Tancik, Jonathan T Barron, Ravi Ramamoorthi, and Ren Ng. Nerf: Representing scenes as neural radiance fields for view synthesis. In *Proc. ECCV*, 2020.
- Nicolas Moenne-Loccoz, Ashkan Mirzaei, Or Perel, Riccardo de Lutio, Janick Martinez Esturo, Gavriel State, Sanja Fidler, Nicholas Sharp, and Zan Gojcic. 3d gaussian ray tracing: Fast tracing of particle scenes. *ACM Transactions on Graphics and SIGGRAPH Asia*, 2024.
- OpenAI. Chatgpt (gpt-5), 2025. Large language model. Available at <https://chat.openai.com/>.
- Roy Or-El, Xuan Luo, Mengyi Shan, Eli Shechtman, Jeong Joon Park, and Ira Kemelmacher-Shlizerman. StyleSDF: High-resolution 3D-consistent image and geometry generation. In *Proc. CVPR*, 2022.

-
- Zijie Pan, Zeyu Yang, Xiatian Zhu, and Li Zhang. Fast dynamic 3d object generation from a single-view video. *arXiv preprint arXiv:2401.08742*, 2024.
- Jangho Park, Taesung Kwon, and Jong Chul Ye. Zero4d: Training-free 4d video generation from single video using off-the-shelf video diffusion. *arXiv preprint arXiv:2503.22622*, 2025.
- Ben Poole, Ajay Jain, Jonathan T. Barron, and Ben Mildenhall. DreamFusion: Text-to-3D using 2D diffusion. In *Proc. ICLR*, 2023.
- Guocheng Qian, Junli Cao, Aliaksandr Siarohin, Yash Kant, Chaoyang Wang, Michael Vasilkovsky, Hsin-Ying Lee, Yuwei Fang, Ivan Skorokhodov, Peiye Zhuang, et al. Atom: Amortized text-to-mesh using 2d diffusion. *arXiv preprint arXiv:2402.00867*, 2024a.
- Guocheng Qian, Jinjie Mai, Abdullah Hamdi, Jian Ren, Aliaksandr Siarohin, Bing Li, Hsin-Ying Lee, Ivan Skorokhodov, Peter Wonka, Sergey Tulyakov, et al. Magic123: One image to high-quality 3D object generation using both 2D and 3D diffusion priors. In *Proc. ICLR*, 2024b.
- Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *Proc. ICML*, 2021.
- Jiawei Ren, Liang Pan, Jiaxiang Tang, Chi Zhang, Ang Cao, Gang Zeng, and Ziwei Liu. DreamGaussian4D: Generative 4D Gaussian splatting. *arXiv preprint arXiv:2312.17142*, 2023.
- Jiawei Ren, Kevin Xie, Ashkan Mirzaei, Hanxue Liang, Xiaohui Zeng, Karsten Kreis, Ziwei Liu, Antonio Torralba, Sanja Fidler, Seung Wook Kim, et al. L4gm: Large 4d gaussian reconstruction model. In *Proc. NeurIPS*, 2024a.
- Xuanchi Ren, Jiahui Huang, Xiaohui Zeng, Ken Museth, Sanja Fidler, and Francis Williams. Xcube: Large-scale 3d generative modeling using sparse voxel hierarchies. In *Proc. CVPR*, 2024b.
- Xuanchi Ren, Yifan Lu, Hanxue Liang, Zhangjie Wu, Huan Ling, Mike Chen, Sanja Fidler, Francis Williams, and Jiahui Huang. Scube: Instant large-scale scene reconstruction using voxplats. *Proc. NeurIPS*, 2024c.
- Xuanchi Ren, Tianchang Shen, Jiahui Huang, Huan Ling, Yifan Lu, Merlin Nimier-David, Thomas Müller, Alexander Keller, Sanja Fidler, and Jun Gao. Gen3c: 3d-informed world-consistent video generation with precise camera control. *Proc. CVPR*, 2025.
- Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *Proc. CVPR*, 2022.
- Aditya Sanghi, Hang Chu, Joseph G Lambourne, Ye Wang, Chin-Yi Cheng, Marco Fumero, and Kamal Rahimi Malekshan. CLIP-Forge: Towards zero-shot text-to-shape generation. In *Proc. CVPR*, 2022.
- Kyle Sargent, Zizhang Li, Tanmay Shah, Charles Herrmann, Hong-Xing Yu, Yunzhi Zhang, Eric Ryan Chan, Dmitry Lagun, Li Fei-Fei, Deqing Sun, et al. Zeronvs: Zero-shot 360-degree view synthesis from a single image. In *Proc. CVPR*, 2024.
- Manuel-Andreas Schneider, Lukas Höllein, and Matthias Nießner. Worldexplorer: Towards generating fully navigable 3d scenes. *arXiv preprint arXiv:2506.01799*, 2025.
- Katja Schwarz, Axel Sauer, Michael Niemeyer, Yiyi Liao, and Andreas Geiger. VoxGRAF: Fast 3D-aware image synthesis with sparse voxel grids. In *Proc. NeurIPS*, 2022.
- Katja Schwarz, Norman Mueller, and Peter Kotschieder. Generative gaussian splatting: Generating 3d scenes with video diffusion priors. *arXiv preprint arXiv:2503.13272*, 2025.
- Junyoung Seo, Jisang Han, Jaewoo Jung, Siyoon Jin, Jounghbin Lee, Takuya Narihira, Kazumi Fukuda, Takashi Shibuya, Donghoon Ahn, Shoukang Hu, et al. Vid-camedit: Video camera trajectory editing with generative rendering from estimated geometry. *arXiv preprint arXiv:2506.13697*, 2025.

918 Yichun Shi, Peng Wang, Jianglong Ye, Long Mai, Kejie Li, and Xiao Yang. MVDream: Multi-view
919 diffusion for 3D generation. In *Proc. ICLR*, 2024.

920

921 Jaidev Shriram, Alex Trevithick, Lingjie Liu, and Ravi Ramamoorthi. Realmdreamer: Text-driven 3d
922 scene generation with inpainting and depth diffusion. *arXiv preprint arXiv:2404.07199*, 2024.

923

924 Karen Simonyan and Andrew Zisserman. Very deep convolutional networks for large-scale image
925 recognition. *arXiv preprint arXiv:1409.1556*, 2014.

926

927 Uriel Singer, Shelly Sheynin, Adam Polyak, Oron Ashual, Iurii Makarov, Filippos Kokkinos, Naman
928 Goyal, Andrea Vedaldi, Devi Parikh, Justin Johnson, et al. Text-to-4d dynamic scene generation.
929 In *Proc. ICML*, 2023.

930

931 Jingxiang Sun, Bo Zhang, Ruizhi Shao, Lizhen Wang, Wen Liu, Zhenda Xie, and Yebin Liu.
932 DreamCraft3D: Hierarchical 3D generation with bootstrapped diffusion prior. In *Proc. ICLR*,
933 2024a.

934

935 Keqiang Sun, Dor Litvak, Yunzhi Zhang, Hongsheng Li, Jiajun Wu, and Shangzhe Wu. Ponymation:
936 Learning articulated 3d animal motions from unlabeled online videos. In *Proc. ECCV*, 2024b.

937

938 Qi Sun, Zhiyang Guo, Ziyu Wan, Jing Nathan Yan, Shengming Yin, Wengang Zhou, Jing Liao, and
939 Houqiang Li. Eg4d: Explicit generation of 4d object without score distillation. *arXiv preprint*
940 *arXiv:2405.18132*, 2024c.

941

942 Wenqiang Sun, Shuo Chen, Fangfu Liu, Zilong Chen, Yueqi Duan, Jun Zhang, and Yikai Wang.
943 Dimensionx: Create any 3d and 4d scenes from a single image with controllable video diffusion.
944 *arXiv preprint arXiv:2411.04928*, 2024d.

945

946 Stanislaw Szymanowicz, Christian Rupprecht, and Andrea Vedaldi. Viewset diffusion:(0-) image-
947 conditioned 3d generative models from 2d data. In *Proc. ICCV*, 2023.

948

949 Stanislaw Szymanowicz, Christian Rupprecht, and Andrea Vedaldi. Splatter image: Ultra-fast
950 single-view 3D reconstruction. In *Proc. CVPR*, 2024.

951

952 Stanislaw Szymanowicz, Eldar Insafutdinov, Chuanxia Zheng, Dylan Campbell, João F Henriques,
953 Christian Rupprecht, and Andrea Vedaldi. Flash3d: Feed-forward generalisable 3d scene recon-
954 struction from a single image. *Proc. 3DV*, 2025a.

955

956 Stanislaw Szymanowicz, Jason Y Zhang, Pratul Srinivasan, Ruiqi Gao, Arthur Brussee, Aleksander
957 Holynski, Ricardo Martin-Brualla, Jonathan T Barron, and Philipp Henzler. Bolt3d: Generating 3d
958 scenes in seconds. *arXiv preprint arXiv:2503.14445*, 2025b.

959

960 Jiaxiang Tang, Zhaoxi Chen, Xiaokang Chen, Tengfei Wang, Gang Zeng, and Ziwei Liu. LGM:
961 Large multi-view gaussian model for high-resolution 3d content creation. In *Proc. ECCV*, 2024a.

962

963 Jiaxiang Tang, Zhaoxi Chen, Xiaokang Chen, Tengfei Wang, Gang Zeng, and Ziwei Liu. LGM:
964 Large multi-view gaussian model for high-resolution 3d content creation. *Proc. ECCV*, 2024b.

965

966 Junshu Tang, Tengfei Wang, Bo Zhang, Ting Zhang, Ran Yi, Lizhuang Ma, and Dong Chen.
967 Make-it-3D: High-fidelity 3D creation from a single image with diffusion prior. *arXiv preprint*
968 *arXiv:2303.14184*, 2023a.

969

970 Shitao Tang, Fuyang Zhang, Jiacheng Chen, Peng Wang, and Yasutaka Furukawa. Mvdifffusion: En-
971 abling holistic multi-view image generation with correspondence-aware diffusion. *Proc. NeurIPS*,
2023b.

972

973 Zhenggang Tang, Peiye Zhuang, Chaoyang Wang, Aliaksandr Siarohin, Yash Kant, Alexander
974 Schwing, Sergey Tulyakov, and Hsin-Ying Lee. Pixel-aligned multi-view generation with depth
975 guided decoder. *arXiv preprint arXiv:2408.14016*, 2024c.

976

977 Ayush Tewari, Tianwei Yin, George Cazenavette, Semon Rezhchikov, Josh Tenenbaum, Frédo Durand,
978 Bill Freeman, and Vincent Sitzmann. Diffusion with forward models: Solving stochastic inverse
979 problems without direct supervision. In *Proc. NeurIPS*, 2023.

972 Dmitry Tochilkin, David Pankratz, Zexiang Liu, Zixuan Huang, Adam Letts, Yangguang Li, Ding
973 Liang, Christian Laforte, Varun Jampani, and Yan-Pei Cao. Triposr: Fast 3D object reconstruction
974 from a single image. *arXiv preprint arXiv:2403.02151*, 2024.

975 Lukas Uzolas, Elmar Eisemann, and Petr Kellnhofer. Motiondreamer: Zero-shot 3d mesh animation
976 from video diffusion models. *arXiv preprint arXiv:2405.20155*, 2024.

977 Basile Van Hoorick, Rundi Wu, Ege Ozguroglu, Kyle Sargent, Ruoshi Liu, Pavel Tokmakov, Achal
978 Dave, Changxi Zheng, and Carl Vondrick. Generative camera dolly: Extreme monocular dynamic
979 novel view synthesis. In *Proc. ECCV*, 2024.

980 Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz
981 Kaiser, and Illia Polosukhin. Attention is all you need. In *Proc. NeurIPS*, 2017.

982 Vikram Voleti, Chun-Han Yao, Mark Boss, Adam Letts, David Pankratz, Dmitry Tochilkin, Chris-
983 tian Laforte, Robin Rombach, and Varun Jampani. SV3D: Novel multi-view synthesis and 3D
984 generation from a single image using latent video diffusion. *arXiv preprint arXiv:2403.12008*,
985 2024.

986 Team Wan. Wan: Open and advanced large-scale video generative models. *arXiv preprint*
987 *arXiv:2503.20314*, 2025.

988 Ziyu Wan, Despoina Paschalidou, Ian Huang, Hongyu Liu, Bokui Shen, Xiaoyu Xiang, Jing Liao,
989 and Leonidas Guibas. CAD: Photorealistic 3D generation via adversarial distillation. In *Proc.*
990 *CVPR*, 2024.

991 Can Wang, Menglei Chai, Mingming He, Dongdong Chen, and Jing Liao. Clip-NeRF: Text-and-image
992 driven manipulation of neural radiance fields. In *Proc. CVPR*, 2022.

993 Chaoyang Wang, Peiye Zhuang, Tuan Duc Ngo, Willi Menapace, Aliaksandr Siarohin, Michael
994 Vasilkovsky, Ivan Skorokhodov, Sergey Tulyakov, Peter Wonka, and Hsin-Ying Lee. 4real-video:
995 Learning generalizable photo-realistic 4d video diffusion. *Proc. CVPR*, 2025a.

996 Haochen Wang, Xiaodan Du, Jiahao Li, Raymond A Yeh, and Greg Shakhnarovich. Score Jacobian
997 chaining: Lifting pretrained 2d diffusion models for 3D generation. In *Proc. CVPR*, 2023a.

998 Peng Wang and Yichun Shi. Imagedream: Image-prompt multi-view diffusion for 3d generation.
999 *arXiv preprint arXiv:2312.02201*, 2023.

1000 Qinghe Wang, Yawen Luo, Xiaoyu Shi, Xu Jia, Huchuan Lu, Tianfan Xue, Xintao Wang, Pengfei
1001 Wan, Di Zhang, and Kun Gai. Cinemaster: A 3d-aware and controllable framework for cinematic
1002 text-to-video generation. In *SIGGRAPH*, 2025b.

1003 Ruicheng Wang, Sicheng Xu, Cassie Dai, Jianfeng Xiang, Yu Deng, Xin Tong, and Jiaolong Yang.
1004 Moge: Unlocking accurate monocular geometry estimation for open-domain images with optimal
1005 training supervision, 2024a. URL <https://arxiv.org/abs/2410.19115>.

1006 Shuzhe Wang, Vincent Leroy, Yohann Cabon, Boris Chidlovskii, and Jerome Revaud. Dust3r:
1007 Geometric 3d vision made easy. In *Proc. CVPR*, 2024b.

1008 Xi Wang, Robin Courant, Marc Christie, and Vicky Kalogeiton. Akira: Augmentation kit on rays for
1009 optical video generation. In *Proc. CVPR*, 2025c.

1010 Xiaodong Wang, Zhirong Wu, and Peixi Peng. Longdwm: Cross-granularity distillation for building
1011 a long-term driving world model. *arXiv preprint arXiv:2506.01546*, 2025d.

1012 Yikai Wang, Xinzhou Wang, Zilong Chen, Zhengyi Wang, Fuchun Sun, and Jun Zhu. Vidu4d: Single
1013 generated video to high-fidelity 4d reconstruction with dynamic gaussian surfels. *arXiv preprint*
1014 *arXiv:2405.16822*, 2024c.

1015 Yikai Wang, Guangce Liu, Xinzhou Wang, Zilong Chen, Jiafang Li, Xin Liang, Fuchun Sun, and
1016 Jun Zhu. Video4dgen: Enhancing video and 4d generation through mutual optimization. *arXiv*
1017 *preprint arXiv:2504.04153*, 2025e.

1026 Yizhi Wang, Mingrui Zhao, Ali Mahdavi-Amiri, and Hao Zhang. Act-r: Adaptive camera trajectories
1027 for single view 3d reconstruction. *arXiv preprint arXiv:2505.08239*, 2025f.

1028

1029 Yuelei Wang, Jian Zhang, Pengtao Jiang, Hao Zhang, Jinwei Chen, and Bo Li. Cpa: Camera-
1030 pose-awareness diffusion transformer for video generation. *arXiv preprint arXiv:2412.01429*,
1031 2024d.

1032 Zhengyi Wang, Cheng Lu, Yikai Wang, Fan Bao, Chongxuan Li, Hang Su, and Jun Zhu. Prolific-
1033 Dreamer: High-fidelity and diverse text-to-3D generation with variational score distillation. In
1034 *Proc. NeurIPS*, 2023b.

1035

1036 Zhouxia Wang, Ziyang Yuan, Xintao Wang, Tianshui Chen, Menghan Xia, Ping Luo, and Yin Shan.
1037 Motionctrl: A unified and flexible motion controller for video generation. In *SIGGRAPH*, 2024e.

1038 Daniel Watson, William Chan, Ricardo Martin-Brualla, Jonathan Ho, Andrea Tagliasacchi, and
1039 Mohammad Norouzi. Novel view synthesis with diffusion models. *Proc. ICLR*, 2023.

1040

1041 Daniel Watson, Saurabh Saxena, Lala Li, Andrea Tagliasacchi, and David J Fleet. Controlling space
1042 and time with diffusion models. *arXiv preprint arXiv:2407.07860*, 2024.

1043 Qi Wu, Janick Martinez Esturo, Ashkan Mirzaei, Nicolas Moenne-Loccoz, and Zan Gojcic. 3dgut:
1044 Enabling distorted cameras and secondary rays in gaussian splatting. In *Proc. CVPR*, 2025a.

1045

1046 Rundui Wu, Ruiqi Gao, Ben Poole, Alex Trevithick, Changxi Zheng, Jonathan T Barron, and Alek-
1047 sander Holynski. Cat4d: Create anything in 4d with multi-view video diffusion models. *Proc.*
1048 *CVPR*, 2025b.

1049 Tong Wu, Shuai Yang, Ryan Po, Yinghao Xu, Ziwei Liu, Dahua Lin, and Gordon Wetzstein. Video
1050 world models with long-term spatial memory. *arXiv preprint arXiv:2506.05284*, 2025c.

1051

1052 Zijie Wu, Chaohui Yu, Yanqin Jiang, Chenjie Cao, Fan Wang, and Xiang Bai. Sc4d: Sparse-controlled
1053 video-to-4d generation and motion transfer. *arXiv preprint arXiv:2404.03736*, 2024.

1054 Zeqi Xiao, Yifan Zhou, Shuai Yang, and Xingang Pan. Video diffusion models are training-free
1055 motion interpreter and controller. *arXiv preprint arXiv:2405.14864*, 2024.

1056

1057 Kevin Xie, Jonathan Lorraine, Tianshi Cao, Jun Gao, James Lucas, Antonio Torralba, Sanja Fidler,
1058 and Xiaohui Zeng. LATTE3D: Large-scale amortized text-to-enhanced3D synthesis. In *Proc.*
1059 *ECCV*, 2024a.

1060 Yiming Xie, Chun-Han Yao, Vikram Voleti, Huaizu Jiang, and Varun Jampani. Sv4d: Dynamic 3d
1061 content generation with multi-frame and multi-view consistency. *arXiv preprint arXiv:2407.17470*,
1062 2024b.

1063

1064 Jinbo Xing, Long Mai, Cusuh Ham, Jiahui Huang, Aniruddha Mahapatra, Chi-Wing Fu, Tien-Tsin
1065 Wong, and Feng Liu. Motioncanvas: Cinematic shot design with controllable image-to-video
1066 generation. In *SIGGRAPH*, 2025.

1067 Dejia Xu, Yifan Jiang, Chen Huang, Liangchen Song, Thorsten Gernoth, Liangliang Cao, Zhangyang
1068 Wang, and Hao Tang. Cavia: Camera-controllable multi-view video diffusion with view-integrated
1069 attention. *arXiv preprint arXiv:2410.10774*, 2024a.

1070

1071 Dejia Xu, Hanwen Liang, Neel P Bhatt, Hezhen Hu, Hanxue Liang, Konstantinos N Platanotis,
1072 and Zhangyang Wang. Comp4d: Llm-guided compositional 4d scene generation. *arXiv preprint*
1073 *arXiv:2403.16993*, 2024b.

1074 Dejia Xu, Weili Nie, Chao Liu, Sifei Liu, Jan Kautz, Zhangyang Wang, and Arash Vahdat. Camco:
1075 Camera-controllable 3d-consistent image-to-video generation. *arXiv preprint arXiv:2406.02509*,
1076 2024c.

1077

1078 Yinghao Xu, Zifan Shi, Wang Yifan, Hansheng Chen, Ceyuan Yang, Sida Peng, Yujun Shen, and
1079 Gordon Wetzstein. GRM: Large Gaussian reconstruction model for efficient 3D reconstruction
and generation. In *Proc. ECCV*, 2024d.

1080 Yinghao Xu, Hao Tan, Fujun Luan, Sai Bi, Peng Wang, Jiahao Li, Zifan Shi, Kalyan Sunkavalli,
1081 Gordon Wetzstein, Zexiang Xu, et al. DMV3D: Denoising multi-view diffusion using 3D large
1082 reconstruction model. In *Proc. ICLR*, 2024e.

1083 Zhen Xu, Zhengqin Li, Zhao Dong, Xiaowei Zhou, Richard Newcombe, and Zhaoyang Lv.
1084 4dgt: Learning a 4d gaussian transformer using real-world monocular videos. *arXiv preprint*
1085 *arXiv:2506.08015*, 2025.

1086 Zhongcong Xu, Jianfeng Zhang, Jun Hao Liew, Wenqing Zhang, Song Bai, Jiashi Feng, and
1087 Mike Zheng Shou. Pv3d: A 3d generative model for portrait video generation. In *Proc. ICLR*,
1088 2023.

1089 Yunzhi Yan, Zhen Xu, Haotong Lin, Haian Jin, Haoyu Guo, Yida Wang, Kun Zhan, Xianpeng Lang,
1090 Hujun Bao, Xiaowei Zhou, et al. Streetcrafter: Street view synthesis with controllable video
1091 diffusion models. In *Proc. CVPR*, 2025.

1092 Qitong Yang, Mingtao Feng, Zijie Wu, Shijie Sun, Weisheng Dong, Yaonan Wang, and Ajmal Mian.
1093 Beyond skeletons: Integrative latent mapping for coherent 4d sequence generation. *arXiv preprint*
1094 *arXiv:2403.13238*, 2024a.

1095 Zeyu Yang, Zijie Pan, Chun Gu, and Li Zhang. Diffusion²: Dynamic 3d content generation via score
1096 composition of orthogonal diffusion models. *arXiv preprint 2404.02148*, 2024b.

1097 Zhongqi Yang, Wenhong Ge, Yuqi Li, Jiaqi Chen, Haoyuan Li, Mengyin An, Fei Kang, Hua Xue,
1098 Baixin Xu, Yuyang Yin, et al. Matrix-3d: Omnidirectional explorable 3d world generation. *arXiv*
1099 *preprint arXiv:2508.08086*, 2025.

1100 Junliang Ye, Fangfu Liu, Qixiu Li, Zhengyi Wang, Yikai Wang, Xinzhou Wang, Yueqi Duan,
1101 and Jun Zhu. DreamReward: Text-to-3D generation with human preference. *arXiv preprint*
1102 *arXiv:2403.14613*, 2024.

1103 Vickie Ye, Ruilong Li, Justin Kerr, Matias Turkulainen, Brent Yi, Zhuoyang Pan, Otto Seiskari,
1104 Jianbo Ye, Jeffrey Hu, Matthew Tancik, and Angjoo Kanazawa. gsplat: An open-source library for
1105 gaussian splatting. *JMLR*, 2025.

1106 Yuyang Yin, Dejia Xu, Zhangyang Wang, Yao Zhao, and Yunchao Wei. 4dgen: Grounded 4d content
1107 generation with spatial-temporal consistency. *arXiv preprint arXiv:2312.17225*, 2023.

1108 Paul Yoo, Jiaxian Guo, Yutaka Matsuo, and Shixiang Shane Gu. DreamSparse: Escaping from Plato’s
1109 cave with 2D diffusion model given sparse views. In *arXiv preprint arXiv:2306.03414*, 2023.

1110 Heng Yu, Chaoyang Wang, Peiye Zhuang, Willi Menapace, Aliaksandr Siarohin, Junli Cao, Laszlo A
1111 Jeni, Sergey Tulyakov, and Hsin-Ying Lee. 4real: Towards photorealistic 4d scene generation via
1112 video diffusion models. In *Proc. NeurIPS*, 2024a.

1113 Mark YU, Wenbo Hu, Jinbo Xing, and Ying Shan. Trajectorycrafter: Redirecting camera trajectory
1114 for monocular videos via diffusion models. *Proc. ICCV*, 2025.

1115 Wangbo Yu, Jinbo Xing, Li Yuan, Wenbo Hu, Xiaoyu Li, Zhipeng Huang, Xiangjun Gao, Tien-
1116 Tsin Wong, Ying Shan, and Yonghong Tian. Viewcrafter: Taming video diffusion models for
1117 high-fidelity novel view synthesis. *arXiv preprint arXiv:2409.02048*, 2024b.

1118 Xin Yu, Yuan-Chen Guo, Yangguang Li, Ding Liang, Song-Hai Zhang, and Xiaojuan Qi. Text-to-3D
1119 with classifier score distillation. *arXiv preprint arXiv:2310.19415*, 2023.

1120 Yu Yuan, Xijun Wang, Yichen Sheng, Prateek Chennuri, Xingguang Zhang, and Stanley Chan.
1121 Generative photography: Scene-consistent camera control for realistic text-to-image synthesis. In
1122 *Proc. CVPR*, 2025.

1123 Yu-Jie Yuan, Leif Kobbelt, Jiwen Liu, Yuan Zhang, Pengfei Wan, Yu-Kun Lai, and Lin Gao. 4dynamic:
1124 Text-to-4d generation with hybrid priors. *arXiv preprint arXiv:2407.12684*, 2024.

1125 Bohan Zeng, Ling Yang, Siyu Li, Jiaming Liu, Zixiang Zhang, Juanxi Tian, Kaixin Zhu, Yongzhen
1126 Guo, Fu-Yun Wang, Minkai Xu, et al. Trans4d: Realistic geometry-aware transition for composi-
1127 tional text-to-4d synthesis. *arXiv preprint arXiv:2410.07155*, 2024a.

1134 Yifei Zeng, Yanqin Jiang, Siyu Zhu, Yuanxun Lu, Youtian Lin, Hao Zhu, Weiming Hu, Xun
1135 Cao, and Yao Yao. Stag4d: Spatial-temporal anchored generative 4d gaussians. *arXiv preprint*
1136 *arXiv:2403.14939*, 2024b.

1137 Bowen Zhang, Tianyu Yang, Yu Li, Lei Zhang, and Xi Zhao. Compress3D: a compressed latent
1138 space for 3D generation from a single image. *arXiv preprint arXiv:2403.13524*, 2024a.

1140 David Junhao Zhang, Roni Paiss, Shiran Zada, Nikhil Karnad, David E Jacobs, Yael Pritch, Inbar
1141 Mosseri, Mike Zheng Shou, Neal Wadhwa, and Nataniel Ruiz. Recapture: Generative video camera
1142 controls for user-provided videos using masked video fine-tuning. *arXiv preprint arXiv:2411.05003*,
1143 2024b.

1144 Haiyu Zhang, Xinyuan Chen, Yaohui Wang, Xihui Liu, Yunhong Wang, and Yu Qiao. 4diffusion:
1145 Multi-view video diffusion model for 4d generation. *arXiv preprint arXiv:2405.20674*, 2024c.

1147 Hao Zhang, Di Chang, Fang Li, Mohammad Soleymani, and Narendra Ahuja. Magicpose4d: Crafting
1148 articulated models with appearance and motion control. *arXiv preprint arXiv:2405.14017*, 2024d.

1149 Kai Zhang, Sai Bi, Hao Tan, Yuanbo Xiangli, Nanxuan Zhao, Kalyan Sunkavalli, and Zexiang Xu.
1150 Gs-lrm: Large reconstruction model for 3d gaussian splatting. In *Proc. ECCV*, 2024e.

1152 Richard Zhang, Phillip Isola, Alexei A Efros, Eli Shechtman, and Oliver Wang. The unreasonable
1153 effectiveness of deep features as a perceptual metric. In *Proc. CVPR*, 2018.

1154 Songchun Zhang, Huiyao Xu, Sitong Guo, Zhongwei Xie, Hujun Bao, Weiwei Xu, and Changqing
1155 Zou. Spatialcrafter: Unleashing the imagination of video diffusion models for scene reconstruction
1156 from limited observations. *arXiv preprint arXiv:2505.11992*, 2025a.

1158 Tianyuan Zhang, Hong-Xing Yu, Rundi Wu, Brandon Y Feng, Changxi Zheng, Noah Snavely, Jiajun
1159 Wu, and William T Freeman. Physdreamer: Physics-based interaction with 3d objects via video
1160 generation. In *Proc. ECCV*, 2024f.

1162 Yisu Zhang, Chenjie Cao, Chaohui Yu, and Jianke Zhu. Lion-lora: Rethinking lora fusion to unify
1163 controllable spatial and temporal generation for video diffusion. *arXiv preprint arXiv:2507.05678*,
1164 2025b.

1165 Zhichao Zhang, Hui Chen, Jinsheng Deng, Xiaoqing Yin, Xingshen Song, and Ming Xu. Motion4d:
1166 A decoupled pipeline for enhanced text-to-4d generation with optimized motion patterns. *SSRN*,
1167 2024g.

1168 Yuyang Zhao, Zhiwen Yan, Enze Xie, Lanqing Hong, Zhenguo Li, and Gim Hee Lee. Animate124:
1169 Animating one image to 4d dynamic scene. *arXiv preprint arXiv:2311.14603*, 2023.

1171 Yuyang Zhao, Chung-Ching Lin, Kevin Lin, Zhiwen Yan, Linjie Li, Zhengyuan Yang, Jianfeng
1172 Wang, Gim Hee Lee, and Lijuan Wang. Genxd: Generating any 3d and 4d scenes. *arXiv preprint*
1173 *arXiv:2411.02319*, 2024.

1174 Guangcong Zheng, Teng Li, Rui Jiang, Yehao Lu, Tao Wu, and Xi Li. Cami2v: Camera-controlled
1175 image-to-video diffusion model. *arXiv preprint arXiv:2410.15957*, 2024a.

1177 Yufeng Zheng, Xueting Li, Koki Nagano, Sifei Liu, Otmar Hilliges, and Shalini De Mello. A unified
1178 approach for text-and image-guided 4d scene generation. In *Proc. CVPR*, 2024b.

1179 Junwei Zhou, Xueting Li, Lu Qi, and Ming-Hsuan Yang. Coco4d: Comprehensive and complex 4d
1180 scene generation. *arXiv preprint arXiv:2506.19798*, 2025.

1182 Tinghui Zhou, Richard Tucker, John Flynn, Graham Fyffe, and Noah Snavely. Stereo magnification:
1183 Learning view synthesis using multiplane images. In *SIGGRAPH*, 2018.

1184 Hanxin Zhu, Tianyu He, Anni Tang, Junliang Guo, Zhibo Chen, and Jiang Bian. Compositional
1185 3d-aware video generation with llm director. *arXiv preprint arXiv:2409.00558*, 2024.

1187 Hanxin Zhu, Tianyu He, Xiqian Yu, Junliang Guo, Zhibo Chen, and Jiang Bian. Ar4d: Autoregressive
4d generation from monocular videos. *arXiv preprint arXiv:2501.01722*, 2025.

1188 Chen Ziwen, Hao Tan, Kai Zhang, Sai Bi, Fujun Luan, Yicong Hong, Li Fuxin, and Zexiang Xu.
1189 Long-lrm: Long-sequence large reconstruction model for wide-coverage gaussian splats. *arXiv*
1190 *preprint arXiv:2410.12781*, 2024.
1191
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A ADDITIONAL DETAILS OF VIDEO DIFFUSION MODEL

This section provides supplementary details on the GEN3C (Ren et al., 2025) video diffusion backbone introduced in Sec. 3.1, with particular emphasis on the modifications we use to further improve its 3D consistency.

Conservative mask refinement. A principal limitation of GEN3C’s forward warping is background leakage: object regions occluded in the source view, such as the lion body in Fig. 8, may become visible from novel viewpoints while remaining improperly represented in the disocclusion mask. Consequently, the video diffusion model may incompletely synthesize the newly visible foreground, as illustrated in Fig. 8 (second and fourth columns).

To mitigate this failure mode, we adopt a conservative refinement strategy. Because the geometry of occluded regions in $\mathbf{P}^{t,v}$ is unavailable, we assume that such regions may be occupied and derive refined disocclusion masks under that assumption. This procedure enables the video model to reason over potentially visible foreground content that standard point-based rendering can fail to mark.

Our approach constructs a triangular mesh from camera-space points $\mathbf{P}_c^{t,v} = \mathbf{C}^t \mathbf{P}^{t,v}$ by connecting spatially adjacent pixels, following the spirit of Hu et al. (2021):

$$\mathcal{M} = \bigcup_{(u,v)} \{ \triangle(\mathbf{p}_{u,v}, \mathbf{p}_{u+1,v}, \mathbf{p}_{u,v+1}), \triangle(\mathbf{p}_{u+1,v}, \mathbf{p}_{u+1,v+1}, \mathbf{p}_{u,v+1}) \} \quad (2)$$

where $\mathbf{p}_{u,v}$ denotes the 3D point at pixel (u, v) in $\mathbf{P}_c^{t,v}$. This surface mesh serves as an approximate boundary between visible and invisible regions. For each rendered pixel, we compare the standard point-based depth $\mathbf{D}^{t,v}$ with the mesh-interpolated depth $\mathbf{D}_{\mathcal{M}}^{t,v}$ obtained through ray-surface intersection:

$$\mathbf{M}^{t,v}(u, v) = \begin{cases} 0 & \text{if } \mathbf{D}_{\mathcal{M}}^{t,v}(u, v) < \mathbf{D}^{t,v}(u, v) - \epsilon \\ \mathbf{M}_{\text{orig}}^{t,v}(u, v) & \text{otherwise} \end{cases}, \quad (3)$$

where $\mathbf{M}_{\text{orig}}^{t,v}$ is the original disocclusion mask produced by forward warping and ϵ is a small tolerance. When the mesh surface lies closer than the original points, we conservatively mask out the corresponding regions as potential foreground disocclusions. As shown in Fig. 8 (third and fifth columns), this refinement yields more reliable guidance for video generation, particularly under large viewpoint changes.

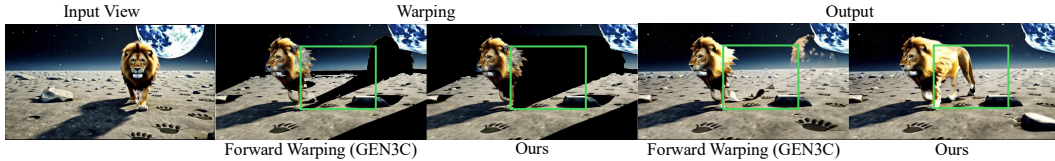


Figure 8: **Rendering function comparison.** Our model builds on GEN3C (Ren et al., 2025), which uses forward-warped images for camera-control conditioning. We compare standard forward warping (left) with our conservative rendering function (right): the highlighted region is incorrectly filled with background pixels under forward warping but is properly masked by our method, enabling correct completion.

B MODEL DETAILS

Progressive training. Directly training the 3DGS decoder on 6 trajectories of 121 frames at 704×1280 resolution is computationally expensive; accordingly, we adopt the progressive training schedule summarized in Tab. 3. We also initialize the dynamic model from the final static pre-trained model. Total training requires 6 days on 8 NVIDIA A100 80 GB GPUs, and we use gsplat (Ye et al., 2025) as the 3DGS implementation.

Subsampling Gaussians. Pixel-aligned 3D Gaussian generation scales linearly with both spatial and temporal resolution. Our generated scenes represent 726 frames at 704×1280 resolution, which would otherwise produce 654,213,120 per-pixel Gaussians. We therefore generate only one

Table 3: **Progressive training setup.** The schedule increases resolution, sequence length, and view coverage across static and dynamic training stages.

Stage	$H \times W$	L	V	S	B	Steps
Static						
1	176×320	17	1	17	4	10k
2	176×320	49	1	49	4	2.5k
3	352×640	49	1	49	2	2.5k
4	704×1280	49	1	49	1	2.5k
5	704×1280	121	1	9	1	57.5k
6	704×1280	121	1-6	9	1	7k
Dynamic						
7	704×1280	121	6	12	1	10k

3D Gaussian per 8×8 spatial neighborhood, reducing the count by a factor of 64 to 10,222,080. Opacity-based pruning further decreases this number to 2,044,416, making the final representation substantially more compact.

Joint Transformer-Mamba blocks. We use the reconstruction block introduced in Long-LRM (Ziwen et al., 2024), in which each block consists of one Transformer layer followed by seven Mamba-2 layers. Repeating this block twice gives 16 layers with 512 hidden dimensions. Notably, this network is considerably smaller than the architectures used in Long-LRM (Ziwen et al., 2024) or Wonderland (Liang et al., 2025a), both of which use 24 layers with 1024 hidden dimensions. Although pure Transformers are often more expressive, they are substantially slower during both training and inference than Mamba2. Because we trained both configurations under the same training budget, we hypothesize that Transformer-only blocks would require many more training iterations to match the rendering quality of the joint design.

Camera encoding. Following prior work (Kant et al., 2024; Zhang et al., 2024e), we encode cameras with Plücker embeddings. These embeddings are computed at full spatial and temporal resolution and then projected into the video latent space with the pre-trained RGB encoder. Since the RGB encoder expects 3 input channels whereas Plücker embeddings are 6-dimensional, we separately encode the 3-dimensional ray directions and the 3-dimensional cross product of ray directions and origins. The resulting latent encodings are concatenated along the channel dimension and mapped to the video latent dimension with a lightweight MLP.

Time encoding. Similarly, to make the 1-dimensional time signal compatible with the 3-channel input expected by the RGB encoder, we concatenate a 2-dimensional sinusoidal embedding to the original time value along the channel dimension. We then replicate the values spatially before encoding time into the video latent space. This procedure permits independent encoding of both source and target times in the compressed latent representation.

Dynamic data augmentation. Fig. 9 visualizes the dynamic data augmentation procedure for two representative camera trajectories. The procedure constructs paired supervision views that observe the same motion state from different extreme viewpoints, thereby improving coverage across time and camera position.

C ADDITIONAL EVALUATION

C.1 ADDITIONAL BASELINE DETAILS

We additionally compare against BTimer (Liang et al., 2025b), a recent method for joint 3D and 4D reconstruction from posed images or videos. We contacted the BTimer authors to conduct experiments on the same scenes used by *Lyra*. Because BTimer is purely regression-based, we integrate it with the GEN3C (Ren et al., 2025) video diffusion backbone. Specifically, we use the same camera trajectories as our model and provide RGB-decoded videos from GEN3C as BTimer inputs. BTimer operates in pixel space and requires 12 input images; however, as discussed in the main paper, pixel-space methods such as BTimer run out of memory when directly applied to the 726 high-resolution frames generated by GEN3C. To make the comparison feasible, we uniformly subsample GEN3C frames. For static evaluation, we select 12 frames that evenly cover the available viewpoints. For dynamic

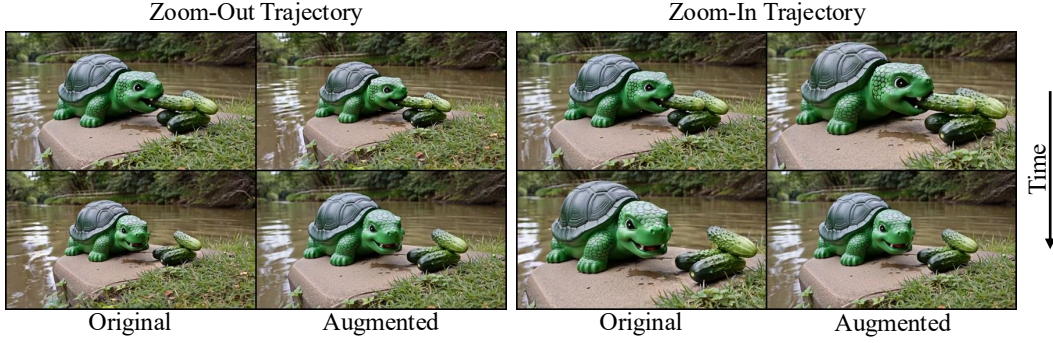


Figure 9: **Dynamic data augmentation videos.** We augment supervision with a motion-reversed video so that each timestep is observed with full spatial coverage, preventing low-opacity artifacts at early timesteps. Two representative trajectories, zoom-out and zoom-in, are shown together with their augmented counterparts, whose camera motion is temporally flipped.



Figure 10: **Image-to-3DGS Comparison.** We compare our method with BTimer (GEN3C) by visualizing five viewpoints from generated scenes, where our reconstructions exhibit markedly fewer artifacts and higher visual fidelity.

evaluation, we sample 11 frames spanning the viewpoint range and additionally include the bullet-time frame corresponding to the motion being reconstructed, ensuring that the target motion state is always available as a reference. We refer to this integrated baseline as BTimer (GEN3C), namely BTimer combined with GEN3C.

C.2 ADDITIONAL STATIC 3D EVALUATION

For image-to-3DGS generation, Fig. 10 presents qualitative comparisons with BTimer (GEN3C), and Tab. 4a reports the corresponding quantitative results. Our method exhibits higher fidelity and fewer artifacts.

C.3 DYNAMIC 3D EVALUATION

BTimer (GEN3C) comparison. For dynamic 3D evaluation, we compare our approach with BTimer (GEN3C), described in Sec. C.1, on 100 out-of-distribution videos from the dynamic *Lyra*

Table 4: Quantitative results on the static and dynamic *Lyra* datasets.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
BTimer (GEN3C)	16.32	0.580	0.427	BTimer (GEN3C)	20.29	0.687	0.315
Ours	24.92	0.834	0.183	Ours	23.07	0.779	0.231
(a) Comparison on the static <i>Lyra</i> dataset.				(b) Comparison on the dynamic <i>Lyra</i> dataset.			

dataset. To ensure fairness, we crop videos to 512×512 , the resolution on which BTimer was primarily trained. As reported in Tab. 4b, our method substantially outperforms BTimer (GEN3C).

C.4 DEPTH VISUALIZATION

Fig. 11 visualizes depth maps produced by our generated 3DGS. Depth supervision prevents flattened geometries while preserving visual quality.

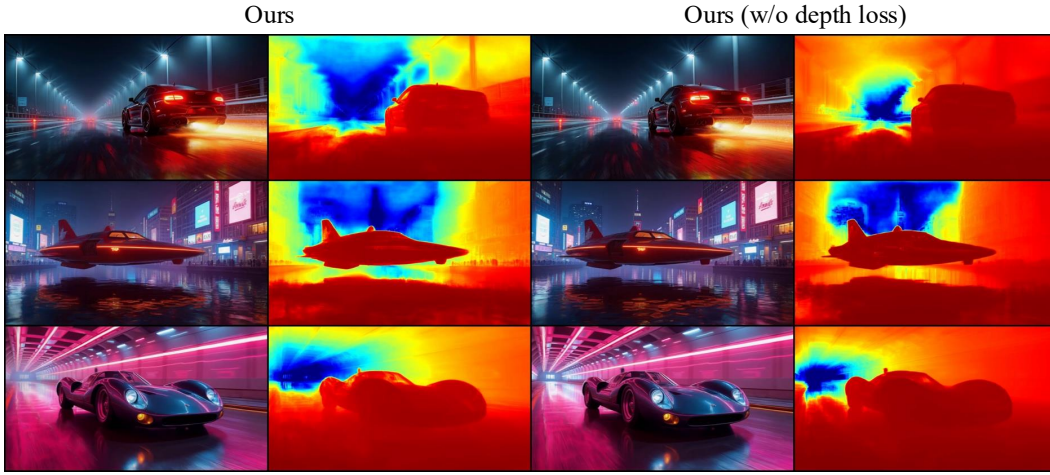


Figure 11: **Depth loss ablation.** We visualize 3DGS renderings together with their corresponding depth maps, showing that depth supervision prevents flattened geometries without degrading visual quality.

D ROBOT SIMULATION

A primary motivation for constructing a diverse 3D generator is its capacity to synthesize simulation environments for autonomous agents. We therefore develop an end-to-end pipeline: first, the model generates 3D scenes from text; second, the predicted 3D Gaussians are exported as a .ply file; and third, the .ply files are converted into .usdz format. We use 3DGUT (Wu et al., 2025a) and its export function to create the .usdz files. The resulting .usdz asset can then be imported into the NVIDIA Isaac robot development platform as a physically based virtual environment. AI-based robots can be trained and tested across diverse conditions in these environments. Early demo results will be released upon acceptance, and Fig. 12 shows a screenshot of the interface.

E RELATED WORK

This section provides additional related-work context. Fig. 13 highlights how our formulation differs from CAT3D (Gao et al., 2024b) and Bolt3D (Szymanowicz et al., 2025b).

3D generation. Early 3D generation methods primarily targeted single object categories, extending GANs into the 3D domain by using neural renderers as inductive biases (DeVries et al., 2021; Chan et al., 2022; Or-El et al., 2022; Schwarz et al., 2022; Bahmani et al., 2023b). With the introduction of CLIP-based supervision (Radford et al., 2021), the field moved toward more flexible and diverse asset

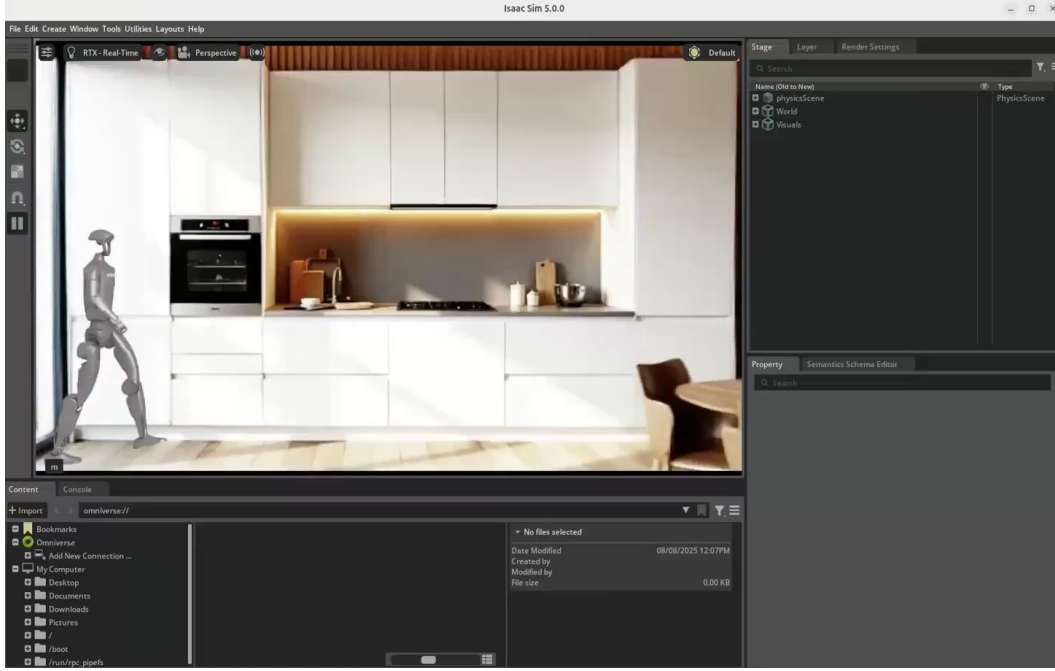


Figure 12: **Robot Simulation.** We show a frame from a robot simulation in Isaac Sim 5.0, where the simulation environment is instantiated from a 3DGS scene generated by our method.

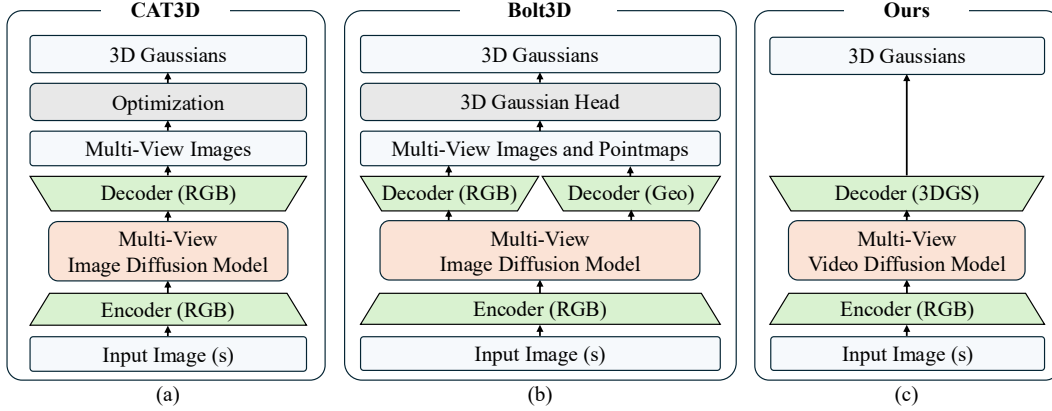


Figure 13: **Approaches for 3D Generation.** CAT3D (Gao et al., 2024b) introduces a multi-view image diffusion model that synthesizes images from novel viewpoints, after which those images are reconstructed into 3D Gaussians through optimization. Bolt3D (Szymanowicz et al., 2025b) fine-tunes CAT3D to output pointmaps via a geometry autoencoder and replaces optimization with a feed-forward 3D Gaussian head. By contrast, our method builds on a pre-trained camera-controlled video diffusion model and *directly* decodes multi-view video latents into 3D Gaussians.

creation, enabling both text-driven generation and editing (Chen et al., 2018; Jain et al., 2022; Sanghi et al., 2022; Jetchev, 2021; Gao et al., 2023; Wang et al., 2022). More recently, diffusion models have substantially improved generation quality by replacing CLIP guidance with Score Distillation Sampling (SDS) (Poole et al., 2023; Wang et al., 2023b; Lin et al., 2023a; Chen et al., 2023b; Liang et al., 2023; Wang et al., 2023a; Li et al., 2024e; He et al., 2024b; Ye et al., 2024; Liu et al., 2024b; Yu et al., 2023; Katzir et al., 2024; Lee et al., 2024a; Sun et al., 2024a).

To improve structural coherence, many methods impose multi-view consistency by generating scenes from multiple perspectives (Lin et al., 2023b; Liu et al., 2023; Shi et al., 2024; Feng et al., 2024a; Liu et al., 2024a; Kim et al., 2023; Voleti et al., 2024; Höllein et al., 2024; Tang et al., 2024c; Gao et al., 2024b; Wang et al., 2025f; Kant et al., 2025; Yuan et al., 2025; Ren et al., 2024b), whereas others

use iterative inpainting for scene synthesis (Höllerin et al., 2023; Shriram et al., 2024). A related line of work lifts 2D images into 3D with NeRF (Mildenhall et al., 2020), 3D Gaussian Splatting (Kerbl et al., 2023), or mesh-based representations combined with diffusion models (Chan et al., 2023; Tang et al., 2023a; Gu et al., 2023; Liu et al., 2024c; Yoo et al., 2023; Tewari et al., 2023; Qian et al., 2024b; Long et al., 2024; Wan et al., 2024; Szymanowicz et al., 2023; Lu et al., 2024).

Feed-forward 3D models. Complementary research studies fast feed-forward techniques that directly predict 3D content from image or text inputs (Hong et al., 2024; Li et al., 2024b; Xu et al., 2024e;d; Zhang et al., 2024a; Han et al., 2024; Jiang & Wang, 2024; Xie et al., 2024a; Tang et al., 2024b; Tochilkin et al., 2024; Qian et al., 2024a; Szymanowicz et al., 2024; 2025a; Liang et al., 2025a; Szymanowicz et al., 2025b; Schwarz et al., 2025; Yang et al., 2025; Zhang et al., 2025a). In contrast to our method, however, these approaches remain limited to static 3D scene generation. Other methods specialize in particular scene classes, such as faces (Kirschstein et al., 2025). Concurrent work (Liang et al., 2025b; Xu et al., 2025) addresses real scenes but cannot handle diverse generated scenes or large viewpoint changes.

4D generation. The last several years have brought rapid progress in 4D generation, or dynamic 3D scene synthesis. Many methods use text prompts or images as guidance. Following the introduction of large-scale generative models for this task (Singer et al., 2023), subsequent work has improved both visual fidelity and motion quality (Ren et al., 2023; Ling et al., 2024a; Bahmani et al., 2024b; Zheng et al., 2024b; Gao et al., 2024a; Yang et al., 2024a; Jiang et al., 2024b; Miao et al., 2024; Li et al., 2024a; Zhang et al., 2024g; Yuan et al., 2024; Jiang et al., 2024a).

Although text conditioning is common, alternative approaches seek to lift 2D images or videos into dynamic 3D scenes (Ren et al., 2023; Zhao et al., 2023; Yin et al., 2023; Pan et al., 2024; Zheng et al., 2024b; Ling et al., 2024a; Gao et al., 2024a; Zeng et al., 2024b; Chu et al., 2024; Wu et al., 2024; Yang et al., 2024b; Wang et al., 2024c; Feng et al., 2024b; Sun et al., 2024c; Zhang et al., 2024c; Ren et al., 2024a; Lee et al., 2024b; Li et al., 2024c; Van Hoorick et al., 2024; Uzolas et al., 2024; Chai et al., 2024; Liang et al., 2024; Li et al., 2024d; Xie et al., 2024b; Li et al., 2025a; Wang et al., 2025e; Zhu et al., 2025; Zhou et al., 2025; Hu et al., 2025; Park et al., 2025). Additional efforts incorporate physics priors into generation pipelines (Lin et al., 2024; Huang et al., 2024b; Zhang et al., 2024f; Kiray et al., 2025) or improve motion controllability with template-driven methods (Zhang et al., 2024d; Sun et al., 2024b; Chen et al., 2024b). A parallel direction emphasizes compositional and interactive 4D generation (Bahmani et al., 2024a; Xu et al., 2024b; Cao et al., 2024; Yu et al., 2024a; Zeng et al., 2024a; Zhu et al., 2024).

Another strand extends 3D GAN frameworks to 4D by training directly on 2D video data (Bahmani et al., 2023a; Xu et al., 2023). Such methods, however, are typically restricted by small-scale, single-category datasets, which limits generalization. In addition, most remain object-centric and often neglect backgrounds. Their overall visual fidelity therefore lags behind the photorealism of recent video generation models, which our method leverages.

Camera-conditioned video models. Recent work has increasingly explored camera control in video diffusion models. MotionCtrl (Wang et al., 2024e) introduced camera conditioning by augmenting pre-trained video models (Chen et al., 2023a; Blattmann et al., 2023) with extrinsic matrices. Subsequent methods (He et al., 2024a; Xu et al., 2024c; Kuang et al., 2024; Ju et al., 2025; He et al., 2025; Li et al., 2025c; Wang et al., 2025c; 2024d; Lei et al., 2025; Liu et al., 2025) strengthened conditioning by representing cameras with Plücker coordinates, while others follow MotionCtrl by using flattened poses (Wang et al., 2025b). Another line of work (Hu et al., 2024; Xiao et al., 2024; Ling et al., 2024c; Hou et al., 2024) enables camera motion control without adding trainable parameters. Despite these advances, the preceding methods rely on U-Net-based architectures. Other approaches investigate low-rank strategies for camera control (Zhang et al., 2025b).

Several recent works continue extending U-Net-based camera-control pipelines (Zhao et al., 2024; Xu et al., 2024a; Zheng et al., 2024a; Zhang et al., 2024b; Yu et al., 2024b; Lei et al., 2024), whereas others begin to explore diffusion transformers. Cheong et al. (Cheong et al., 2024) propose a transformer-based framework, although its scene and visual quality remain limited. DimensionX (Sun et al., 2024d) introduces joint space-time control in diffusion transformers but relies on predefined, non-continuous camera trajectories. Other methods (Huang et al., 2024a; Mai et al., 2025; Jiang et al., 2025b) study pose and geometry estimation with a video DiT. CAT4D (Wu et al., 2025b) proposes a multi-view video diffusion model obtained by fine-tuning a multi-view backbone. Large-

scale synthetic data generation further improves performance, as shown by SynCamMaster (Bai et al., 2025b) and ReCamMaster (Bai et al., 2025a), which achieve strong camera-controlled video generation with transformer-based architectures. Finally, 4DiM (Watson et al., 2024) trains a space-time diffusion model from scratch for single-image novel-view synthesis. Another line of work retargets camera motion in videos containing underlying scene motion (Seo et al., 2025; Ma et al., 2025). A prominent direction uses spatiotemporal caches, such as point clouds or optical flow, to condition camera-controlled video generation (Wu et al., 2025c; Ren et al., 2025; Wang et al., 2025d; Cao et al., 2025; YU et al., 2025; Jin et al., 2025; Xing et al., 2025; Yan et al., 2025; Li et al., 2025b; Gu et al., 2025).